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Integration and comparison of multi-criteria decision making methods in safe route planner

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ABSTRACT

Motor vehicle crashes are a leading cause of death in the U.S. In order to reduce death and serious injury, road and traffic engineers manually evaluate road segments and visualize the safety level of roads. These existing risk maps can be confusing and must be manually interpreted by drivers to find the safest path from a source to a given destination; this can result in ignoring the safety of the routes by drivers. In addition, common navigation systems such as Google Maps and Waze present two or three alternative paths from a source to a given destination based on the travel time and distance. A navigation system is required to take the safety level of the road segments into consideration while suggesting a path. This navigation system needs to acquire knowledge from various sources, a user interface to obtain user preferences, and an inference engine to find the best paths. Such a system can still suggest multiple conflicting paths, such as shortest, fastest and safest paths. This paper presents the addition of a multi-criteria decision-making (MCDM) method, Analytical Hierarchy Process, to a previously designed Safe Route Planner to aid users in choosing the most suitable path among M alternative paths. Different MCDM methods can generate different results while applied to the same problem. There are a few comparative studies to compare the results of different Multi-Criteria Decision-Making (MCDM) methods. Therefore, a particular attention is devoted to comparing the results of five decision-making techniques, namely AHP, Fuzzy AHP, TOPSIS, Fuzzy TOPSIS and PROMETHEE through two real-world case studies. In addition, the comparative studies fail to adequately quantify the results of the MCDM methods; consequently, another aim of this research is to investigate the applicability of Spearman's rank correlation coefficient, Average Overlap and Discounted Cumulative Gain techniques to quantify the results of the MCDM methods.

1. Introduction

Motor vehicle crashes are a leading cause of death in the United States. According to the National Vital Statistics Report (NVSR) (Kochanek, Murphy, Xu, & Arias, 2019), there were 40,231 motor vehicle fatalities in 2017 in the United States, which is a similar trend to 2016 (Xu, Murphy, Kochanek, Bastian, & Arias, 2018), and a 6.5% upward trend in comparison to 2015 (Murphy, Xu, Kochanek, Curtin, & Arias, 2017). Several factors such as lack of experience, speeding, roadway design, vehicle design, distraction, impaired driving, and environmental factors contribute to motor vehicle crashes.

There are roads with design issues or high-risk roads that increase the chance of accident occurrences. Road and traffic engineer experts are constantly evaluating the safety level of road segments to improve road infrastructure. Several tools, namely, EuroRap (EuroRAP's protocols, 2015), usRap (Harwood, Gilmore, Bauer, Souleyrette, & Hans, 2010), SafeRoadMaps (Hilton, Horan, & Schooley, 2009), ICE (Pack, Wongsuphasawat, VanDaniker, & Filippova, 2009), FARS Encyclopedia ("National Highway Traffic Safety Administration," n.d.), MnCMAT (Crash, 2019), and CMAT (Crash Mapping Analysis Tool (CMAT) 2020) are designed to show the location of accidents on maps or generate a heat map of accidents. Although manual road assessment using specially equipped vehicles and trained personnel is more accurate, it takes a long time to evaluate the roads in every city, state, and country. These tools are necessary but not sufficient. The drivers are required to manually interpret the maps to find the safest route from a source to a given destination. Moreover, drivers might not be comfortable driving on some roads due to poor visibility, the narrow width of the road or also the fear of hitting animals that might jump into the road.

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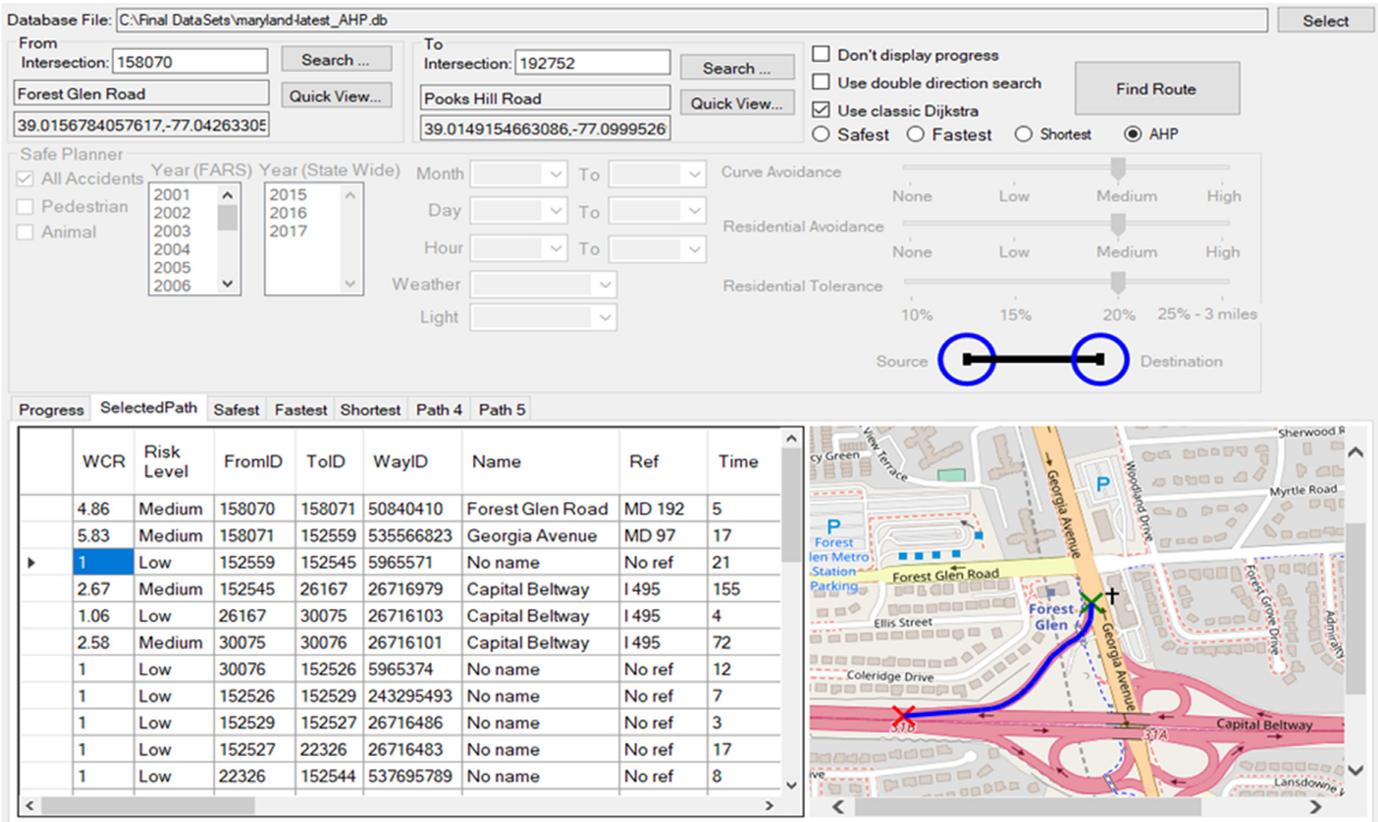

Fig. 1. The user interface of Safe Route Planner.

Current navigation systems such as Google Maps, Apple Maps, Waze and MapQuest suggest the fastest path from a source to a given destination. However, research suggests that drivers prefer a daily commute path that is not necessarily the shortest or fastest path. The preferred path is typically chosen due to the drivers' familiarity with the area, traffic, and physical road conditions. Papinski and Scott (2011) analyzed a real-world GPS data set of 237 home-to-work trips and found that the path drivers take are significantly longer than the shortest distance and fastest path alternatives. Sarraf and McGuire, 2018a, 2018b) presented two different versions of Safe Route Planner. A Safe Route Planner is a navigation system that evaluates the road safety based on historical crash data and a real-time monitoring system, and suggests the safest path from a source to a given destination. As shown in Fig. 1, the Safe Route Planner allows the users to query the historical crashes based on multiple factors such as years, months, days of the week, hours, weather conditions, light conditions, accidents with pedestrians or animals to find the safest route. In addition, the Safe Route Planner has features to avoid curvy roads due to higher probability of accidents and avoid residential roads to keep residential roads safe for residents. The pilot studies proved the concept of detecting high-risk roads and suggesting alternative roads with lower risk of accidents.

The navigation systems usually present multiple alternative paths, and the users must make an uninformed decision to select the best one. The Safe Route Planner adds to this complexity by suggesting a new alternative path that is the safest one. Consequently, users will have even more alternative paths to choose from. The problem is, if the drivers are presented with three different paths, namely fastest, shortest, and safest, from a source to a given destination, which one is the right choice. Different drivers might have different preferences for path selection, and obviously having only one navigation system is not satisfactory. A decision-making system is required to find the priorities of the drivers and assist them in choosing the best path. Hence, a knowledge-based Expert Navigation System needs to be designed to consider the safety of the road based on facts as well as the travel time to suggest the best possible route for individuals.

An Expert System is a fast decision-making system designed to simulate the judgment of an expert and solve complex problems using factual data with high accuracy rate. An expert system has three main components: Knowledge base, inference engine, and user interface. The Expert Navigation System needs to first acquire knowledge from various sources and constantly update its database with new crashes to learn new facts. Second, a user interface is needed to get individual user's preferences based on some predefined criteria and provide interaction between the user and the Expert Navigation. Finally, a forward-chaining inference engine is required to act as an expert to make a decision and find the most suitable path for individuals. In the Expert Navigation System, the knowledge base contains knowledge about road map network, speed limits, degree of road curvature, historical crashes, current crashes, and environmental factors about the crashes such as weather and light conditions. The inference engine is responsible to calculate the probability of accidents under certain conditions for different road segments and suggest the safest path.

It is important to identify and apply a suitable multi-criteria decision-making method to the Safe Route Planner and in general to the transportation field. There are a few comparative studies to compare the results of different Multi-Criteria Decision-Making (MCDM) methods. These comparative studies are mostly related to healthcare and housing (Hodgett, 2016). Different methods can produce different results while applying to an identical problem; therefore, it is important to examine the compatibility of different multi-criteria decision-making methods with a particular type of decision problem (Malczewski & Rinner, 2015). This paper presents the addition of a multi-criteria decision-making system, Analytical Hierarchy Process, to the Safe Route Planner to aid drivers in choosing the best possible path. In addition, the results of five different decision analysis methods, namely, AHP, Fuzzy AHP, TOPSIS, Fuzzy TOPSIS and PROMETHEE are presented.

The existing literature fails to adequately quantify the results of the MCDM methods. Approaches such as Gini Index and Spearman's rank correlation coefficient are not suitable for comparing the ranked list results. In this research, the results of the MCDM methods are evaluated using Spearman's rank correlation coefficient, average overlap and discounted cumulative gain. Next, their limitations are discussed, and the most suitable one is identified.

The main contribution of this paper are as follows:

- Addition of a decision-making method to the Safe Route planner to analyze individual user's judgment based on safety, time and distance criteria.
- Development of a fully automated approach to act as a transportation expert to analyze road safety level and user's judgment to select the most suitable path among M alternative routes from a source to a given destination.
- Comparative study of five multi-criteria decision-making methods, namely AHP, Fuzzy AHP, TOPSIS, Fuzzy TOPSIS and PROMETHEE through two real world case studies.
- Comparison of three approaches to derive fuzzy criteria weights for Fuzzy TOPSIS, and suggesting the only suitable approach.
- Identification of two multi-criteria decision-making methods that produce unreliable results in transportation.
- Comparison of different rank-list evaluation methods, and proposing a suitable method for ranked-list evaluation while comparing MCDA methods.

The paper is organized as follows: In the next section, the Safe Route Planner is briefly explained. In Section 3, the decision-making system is introduced. Section 4 presents the summary use of decision analysis systems in transportation. Section 5 presents the research methodology employed in this research. Sections 6–11 briefly explain each decision analysis method with a simple example, and derives criteria weights which are used in the results section. Sections 12 presents the evaluation methodology. Finally, the results and comparison (Section 12) presents two real case studies, evaluates the judgment of different decision analysis methods, and compares different methods with the predefined baseline.

2. Safe route planner

Safe Route Planner (Sarraf & McGuire, 2018a, 2018b) is a navigation system that directs drivers to drive on roads with lower risk of accidents. It uses historical crash data, namely property damage, injury and fatal crashes, along with real-time crashes, traffic counts, a routing algorithm, and OpenSourceMap (OSM License) to estimate the safety level of the roads based on parameters such as year, month, hour, weather conditions, and light conditions.

First, for each road segment the actual weight (AW) which is the travel time in seconds is calculated as shown in Eq. (1).

$$AW_{(n_i,n_j)} = \left(\frac{Dist\left(n_i,n_j\right)\times 3600}{S\left(n_i,n_j\right)}\right) \tag{1}$$

where n is the node, S is the maximum speed in km/hour, and 3600 is the number of seconds in an hour.

Next, the Safe Route Planner calculates the Weighted Crash Rate (WCR) for each road segment as shown in Eq. (2). Clearly, not all the crashes are the same; therefore, different crash types must be treated differently based on their severity. A weight of two is assigned to every real-time (RTime) crash since the severity is currently unknown, five is assigned to every property damage only (PDO), 15 to injury (Inj) and 30 to fatal crashes. For every road segment, the frequency of each crash type is multiplied by its relevant weight and summed to find the crash rate. The result is first multiplied by 100,000 vehicles traveled, then divided by the number of vehicles that passed the road segment in a year (AADT). Field experts should be able to change the weights of the accidents according to their experience.

$$WCR\left(n_i,n_j\right) = \frac{\begin{pmatrix}\left(Fatal\left(n_i,n_j\right)\times 30\right)+\\ \left(Inj\left(n_i,n_j\right)\times 15\right)+\\ \left(PDO\left(n_i,n_j\right)\times 5\right)+\\ \left(RTime\left(n_i,n_j\right)\times 2\right)\end{pmatrix}\times 100000}{AADT\times 365}+1 \tag{2}$$

The WCR is always a number greater than or equal to one. For roads with no accidents, the WCR equals one. A higher number of accidents on a road results in greater WCR weight, which ultimately results in increasing the weight of the road segment and ignoring that path.

The actual weight of the road segment is multiplied by WCR of the road segment to find the safety weight as in Eq. (3). Finally, the safety weight is used along with Dijkstra algorithm to find the safest path from a source to a given destination.

$$SW\left(n_i,n_j\right) = AW\left(n_i,n_j\right)\ \times WCR\left(n_i,n_j\right) \tag{3}$$

It is important to note that the safest path is a path that is both safe and fast. By replacing the AW in Eq. (3) with the distance, the Safe Route Planner might find even a safer path. However, the travel time will increase.

3. Multi-criteria decision support

Decision makers need to select the optimal option among multiple conflicting criteria. For instance, to buy a car, a decision maker needs to evaluate criteria such as quality, style, reliability, performance and price of the available car models to choose the best compromise alternative car. Multi-criteria decision analysis (MCDA) methods compare the available alternatives against criteria and select the most suitable alternative.

There is no one universal classification of MCDA methods (Sen & Yang, 1998). There are different ways to classify them. One broad classification of MCDA is: 1) Multi-attribute decision-making (MADM). 2) Multi-objective decision-making (MODM). MADM requires a discrete set of predefined alternatives, while MODM is the process of selecting or designing one or more alternatives from an infinite number of continuous possibilities (Colson & De Bruyn, 1989; Zavadskas, Turskis, & Kildiene, 2014).

Roy (1981) presented four basic types of problem formulation: Choice problem formulation to choose the single best alternative. Sorting problem formulation that sorts actions or alternatives into categories. Ordering/ranking problem formulation that puts the alternatives in decreasing preference order. Description problem formulation helps to describe the alternatives and/or their consequences.

Belton and Stewart (2002) classified MCDA methods into three broad categories:

- Value measurement models in which numerical scores are assigned to alternatives in order to rank them.
- Goal, aspiration and reference level models: For each of the criteria a satisfactory level of achievement or goal is established. The method seeks to discover the alternative that is closest to the goal.
- Outranking models (the French school): For each of the criteria, the alternatives are compared pairwise to identify the strength of preferring one to another.

In this research, the authors chose AHP and Fuzzy AHP as value measurement methods, TOPSIS and Fuzzy TOPSIS as goal, aspiration and reference level models, and PROMETHEE as an outranking method. Different methods can generate different results while applied to the same problem (Zanakis, Solomon, Wishart, & Dublish, 1998). Zanakis et al. (1998) stated that it is impossible or difficult to answer questions such as: which method is more appropriate for what type of problem, and what are the advantages and disadvantages of using one method over another.

The Analytical Hierarchy Process (AHP) is a Multi-Criteria Decision Making system (MCDM) developed by Saaty (1980). It was developed to consider both tangible and intangible factors to make the best decision through pairwise comparisons. A bibliometric study of MCDM/ Multi Attribute Utility Theory (MAUT) fields reveals that AHP has the greatest number of scientific publications compared to other MCDM methods (Wallenius et al., 2008). AHP is the most widely applied decision-making system, and it has been used by many organizations, including Xerox, NASA, Air Force Medical Services, IBM and Department of Defense (Forman, 2001).

Fuzzy logic is first introduced by Zadeh (1965) to deal with uncertainties and imprecise inputs. Pedrycz, Ekel, and Parreiras (2011) name fuzzy sets as one of the most fundamental and influential notions in science and engineering. Fuzzy sets bridge a gap in decision making in classical environment and fuzzy environment. In traditional AHP, the decision maker's preference is expressed by exact values on a scale of 1 to 9. However, in reality, human judgment can be vague and imprecise; therefore, it should not be expressed with exact values. Fuzzy AHP is the approach to deal with the vagueness. Fuzzy AHP was first proposed by Van Laarhoven and Pedrycz (1983). There are many different fuzzy AHP methods to derive the fuzzy weights and rank the alternatives (Buckley, 1985; Chang, 1996; (Van Laarhoven and Pedrycz, 1983) Van Laarhoven 1983; (Boender et al., 1989); (Csutora and Buckley, 2001); Mikhailov, 2003; (Wang and Chin, 2011)).

TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution) was first developed by Hwang and Yoon (1981) to determine the best alternative based on both minimization of Euclidean distance from the positive ideal solution and maximization of Euclidean distance from the negative ideal solution. There are several fuzzy TOPSIS methods developed to address the vagueness of the judgment (Abo-Sinna & Abou-El-Enien, 2005; Chen, 2000; Chen & Hwang, 1992; Liang, 1999; Wang & Elhag, 2006; Wang & lee, 2009). A bibliometric-based survey by Zyoud and Fuchs-Hanusch (2017) found that Chen's (2000) fuzzy TOPSIS method is the most cited article in TOPSIS application. Chen's (2000) fuzzy TOPSIS method is applied in this research to calculate the priority weight of alternatives. The rates of alternatives and the importance weight of each criterion are described by linguistic terms, which in turn they can be expressed in triangular fuzzy numbers. Next, a vertex method is defined to measure the distance between two triangular fuzzy numbers.

PROMETHEE (Preference Ranking organization METHod of Enrichment Evaluation) which is an outranking class of multi-criteria decision analysis method was developed by Brans and Vincke (1985). Different versions of PROMETHEE were introduced (Brans & Mareschal, 1992, 1994, 1996, 2005). PROMETHEE I is based on partial ranking, while PROMETHEE II gives a complete ranking of alternatives that allows the decision makers to select the best compromise alternative.

Kahraman, Onar, and Oztaysi (2015) analyzed the publishing frequency of the most used fuzzy MCDM methods from 1980 to 2014 and presented the results using tabular and graphical illustrations. Table 1 shows the number of published papers using SCOPUS database. Based on the results, fuzzy AHP and fuzzy TOPSIS are the first and the second most widely used fuzzy MCDM techniques. The study shows an exponential trend to use fuzzy MCDM methods and suggests that fuzzy AHP and fuzzy TOPSIS methods will continue to be the most used methods in the future.

The rationale for the selection of MCDM methods in this study was as follows: AHP was chosen due to its simplicity, the ability to handle both qualitative and quantitative data, the ability to derive criteria weights, and above all it is the most widely used method in the literature. Therefore, AHP was used as a baseline for the comparison of other decision-making methods. Moreover, most of the MCDM methods depend on other algorithms to derive the criteria weights; consequently, the criteria weights obtained from AHP were fed to other selected methods. Fuzzy AHP was chosen because it can handle vagueness and uncertainty, and it is the most widely used fuzzy MCDM method. Fuzzy AHP considers the uncertainty of human judgement while comparing safety, time and distance against each other to derive the criteria weights. TOPSIS was chosen because it represents human rationale in judgement and tries to minimize the distance to the best solution and maximize the distance from the worst solution. Since in the Safe Route Planner, the best and the worst alternatives for each individual criterion (time, distance and safety) is known, TOPSIS was assumed to demonstrate a promising result and its results must be investigated in this paper. Fuzzy TOPSIS has the same advantage as TOPSIS and it can deal with ambiguity. In addition, fuzzy TOPSIS is the second most widely used fuzzy MCDM method and its results can be compared with fuzzy AHP. PROMETHEE was chosen since it is based on pairwise comparison that comes naturally and it can give a full ranking of alternatives.

One of the shortcomings of MCDM methods is rank reversal which was first observed by Belton & Gear (1983). Rank reversal refers to changes in the ranking of alternatives by addition or deletion of an alternative. Saaty (1987) showed that with absolute measurement, rank always is preserved, while with relative measurement, rank changes. Since Safe Route Planner is based on absolute measurements, it will not suffer from rank reversal.

There are a few studies that compare the results of different decision-making methods. A study performed by Hodgett (2016) indicates that the majority of MCDM related literature just focuses on one decision-making method with a single problem. Hodgett (2016) compared the results of MARE, AHP, and ELECTRE III with respect to equipment selection problem, and encourages the comparison of other decision-making methods such as ANP, TOPSIS and PROMETHEE, in order to examine the compatibility of a wider range of MCDM methods.

Selmi, Kormi, and Ali (2016)) compared the results of ELECTRE III, PROMETHEE I and II, TOPSIS, AHP and PEG through two multi-criteria case studies. The first case study had 10 alternatives and two criteria, the results showed an important similarity between PROMETHEE II and AHP, and the highest-ranking differences were observed between TOPSIS and ELECTREE III methods. Both AHP and PROMETHEE II agree on the ranking of nine alternatives, while there is only one rank agreement between TOPSIS and AHP. The second case study contained five alternatives and eight criteria. The results showed that ELECTREE III, PROMETHEE, and AHP all agreed on alternative one as the best alternative, while TOPSIS and PEG indicated alternative three is the best one. The study employed Gini Index to quantify the rankings dispersion of the decision-making methods and uses the mean value of dispersion to perform rankings comparisons.

The shortcoming of the Selmi et al. (2016)'s approach is that it does not consider that the top of the ranked list is more important than the bottom of the list. As a result, their comparison between AHP-PROMETHEE II and TOPSIS-PROMETHEE II yields the same mean value of the Gini index. AHP-PROMETHEE II switches one item pair at the bottom of the ranked list, while TOPSIS-PROMETHEE II switches one item pair at the top of the list. A change in the ranks at the top of the rank list must have a higher

Table 1
Published papers using fuzzy MCDM methods from 1980 to 2014.

Classification	Method name	Number of publications
Fuzzy Multi Attribute Decision Making (FMADM)	Fuzzy AHP	8284
	Fuzzy TOPSIS	4010
	Fuzzy ANP	1542
	Fuzzy ELECTRE	1153
	Fuzzy VIKOR	964
	Fuzzy DEMATEL	600
	Fuzzy MACBETH	199
	Fuzzy PROMETHEE	Not Available
Fuzzy Multi Objective Decision-Making (MODM)	Fuzzy Multi Objective Linear Programming	1651
	Fuzzy Multi Objective Goal Programming	
	Fuzzy Heuristic MODM	

negative impact on the evaluation. The evaluation method proposed in our research fixes this issue. In addition, we show the step by step calculation of decision-making methods through a numerical example along with the methods used to derive criteria weights; therefore, this can be used as a guide for the readers to learn and apply these methods

Tscheikner-Gratl, Egger, Rauch, and Kleidorfer (2017) compared the results of ELECTRE, AHP, WSM, TOPSIS, and PROMETHEE III for the application in an integrated rehabilitation management scheme using a real-world case study. For this comparison, three different networks, sewer, water and gas, and road were considered. AHP criteria weights were used for the ELECTREE, PROMETHEE III and TOPSIS methods to rank alternatives. As for the sewer network, the results of the AHP and TOPSIS showed the largest similarities followed by WSM, PROMETHEE and finally ELECTREE. For the water distribution network, AHP and PROMETHEE III showed the highest level of similarities followed by TOPSIS, WSM and ELECTREE. Gas distribution network also showed a high similarity between AHP and PROMETHEE, followed by WSM, TOPSIS and ELECTREE. In this study, the standard deviation is used to compare different MCDA methods and again it does not consider that the top of the ranked list is more important than the bottom of the list.

Soloukdar and Parpanchi (2015) applied Fuzzy AHP and fuzzy TOPSIS to rank six business intelligence vendors. There are only two rank agreement between Fuzzy AHP and fuzzy TOPSIS. The research concluded that fuzzy AHP provides more acceptable rankings. Spearman's rank correlation coefficient was used as a method of comparison. While comparing ranked lists, it is important to place higher emphasis on the top of the list. Spearman's rank correlation treats the changes at the top and at the bottom of the ranked list equally; therefore, it cannot be considered as a good comparison method.

Karimi, Mehrdadi, Hashemian, Nabi-Bidhendi, and Tavakkoli-Moghaddam (2011) used fuzzy AHP and fuzzy TOPSIS methods to rank five wastewater treatment processes. Although the ranking results are close to each other, there is only one rank agreement between these two MCDA methods. In contrast, Ertugrul and Karakasoglu (2008) applied fuzzy AHP and fuzzy TOPSIS to three alternatives for facility location selection and the results of both decision-making methods were identical. Similarly, Junior, Osiro, and Carpinetti (2014) applied the Fuzzy AHP and the Fuzzy TOPSIS methods to the problem of supplier selection and compared them in regard to seven factors, namely adequacy to changes of alternatives, adequacy to changes of criteria, agility in the decision process, time complexity, support to group decision making, number of criteria and alternative suppliers, and modeling of uncertainty. Although both decision-making methods resulted in identical ranked lists, they did not suggest a method to compare the resulting ranked lists.

4. MCDA methods in transportation

Multi-criteria decision analysis has been widely used in transportation research. In this section, existing research related to the use of multi-criteria decision making in route planning and transportation are briefly explained.

Analytical Hierarchy Process (AHP) is one of the most widely used decision making system in transportation. Saaty (1995) showed five examples of using AHP in transportation. Saaty's commuter route selection example illustrates the theoretical use of AHP in choosing the best route to commute from Saaty's home to the University of Pennsylvania. Bottlenecks, parking problems, travel time, slowpokes, unreliability, insecurity and fatigue are the used criteria for this route selection. Saaty states that commuters' preferred routes are not necessarily the shortest; some factors such as congestion, light, safety and potholes could influence the route choice.

Niaraki and Kim (2009) used a generic ontology-based architecture and AHP to design a personalized route planning system. Many criteria such as facilities (services, gas station, and terminal), weather, traffic, safety (parking lot, telephone booth, medical center, police station, etc.), road length, and tourist attraction have been utilized in this study.

Yang and Li (2010) presented an enhanced emergency routing method for fire forces based on Dijkstra's algorithm and AHP. Seven impedance factors, namely road length, road width, road type, traffic volume, mass density, velocity limit, junction delay that influence the routing and cause delay were analyzed by AHP to determine the relative weight of each factor and the result is combined with Dijkstra's algorithm to determine the optimal route with minimum travel time.

Choosumrong et al. (2012) and Choosumrong et al. (2014) implemented dynamic routing systems for emergency routing decision planning with the use of AHP. They demonstrated their system in two different scenarios: routing for medical emergency and routing for evacuation. In the routing for medical emergency, they consider the patient's state, availability of beds in nearby hospitals and traffic density. If the nearest hospitals have no available beds and the patient is not in critical condition, the dynamic routing system will show the route to other hospitals. The routing for evacuation calculates the shortest path and avoids blocked areas due to natural disasters such as a flood.

Ghaderi and Pahlavani (2015) integrated simulate annealing algorithm and fuzzy AHP to find optimal multimodal route from a source to a given destination. They used fuzzy AHP to determine the weight of criteria among time, length, fare, and transportation mode preference (subway, bus rapid transit (BRT), taxi, walking, and bus) across route.

Agarwal, Patil, and Mehar (2013) presented a methodology based on AHP for ranking road safety hazardous locations in the absence of crash data. This study considered safety hazardous condition at straight road sections, curve sections, and intersections. Using AHP approach the safety factors such as geometrical condition, surface condition, street light condition and traffic signal condition are rated to determine their relative importance weights. Finally, safety hazardous index was developed using weight of the safety factors.

Kasemsuppakorn and Karimi (2009) presented a personalized routing method suitable for wheelchair users by considering the safety of the sidewalks. They utilized fuzzy logic to express the impedance level for each sidewalk segment, and AHP to derive a numerical weight for each sidewalk parameter such as width, length, slope, sidewalk surface, steps, and sidewalk conditions (cracks, uneven surfaces and manhole covers). Dijkstra's algorithm was used to find the shortest, second shortest and third shortest routes between a given source and a destination. Next, based on the calculated impedance level for each route, the route with the lowest impedance score is selected as the optimal route.

Bozkurt, Yazici, and Keskin (2012) presented a multi criteria route planning by considering time against safety. The safety level used in this paper is solely based on physical road characteristics such as number of lanes, road width, road slope, road surface type and weather condition. These parameters change between 1 and 10 and the result is normalized. The research paper does not specify the exact calculation method of safety degree. The resulting safety level is normalized and Analytical Hierarchical Process and regular increasing monotone are used along with Dijkstra's algorithm to find the preferred path.

Bao, Ruan, Shen, Hermans, and Janssens (2012) incorporated fuzzy TOPSIS to evaluate road safety performance of 21 European countries. Eleven safety performance indicators such as alcohol and drugs, speed, seatbelt wearing rate, vehicle distribution age, road length, motorways density and trauma management were used for road safety evaluation and the number of road fatalities per million inhabitants were used to confirm the results.

Chen, Wang, and Lin (2008) incorporated multi-criteria decision analysis with a routing algorithm for safe transportation of nuclear waste. In this study, AHP was used to calculate the weight of three objectives, namely, minimizing the total transportation cost, minimizing the risk cost, and minimizing the public resistance. The global weights of objectives were used along with Dijkstra's algorithm to find the best suitable path.

Li, Anavatti, and Ray (2014) used Fuzzy AHP to find the optimum route in traffic conditions. They considered travel time, travel distance and traffic density as the three main criteria for k routes. Gumus (2009) employed a two-step fuzzy-AHP and TOPSIS methodology to evaluate hazardous waste transportation firms. Awasthi, Chauhan, and Omrani (2011) utilized fuzzy TOPSIS in evaluating sustainable transportation systems. Anagnostopoulos, Giannoupoulo, and Roukounis (2003) employed PROMETHEE and GAIA methods to evaluate the priorities land transportation infrastructure projects.

The existing research in this area lacks one or more of the followings:

- Some transportation research solely analyze the safety of the roads. They lack to implement a routing algorithm to offer a navigation system.
- Dynamic routing approaches such as personalized navigation systems lack to consider safety of the roads.
- Navigation systems that consider safety do not incorporate real crash data. Factors such as physical road characteristics, availability of hospitals, police stations are considered as safety factors. In addition, safety level of the roads might change due to the changes in the design of the roads; therefore, a real-time monitoring system is required to constantly update the safety level of the roads.
- The road network and the alternative paths are manually selected, and the characteristics of the roads are manually measured.
- Only one decision analysis approach is considered; therefore, it is not clear which decision analysis approach outperforms other methods.
- No suitable evaluation method is suggested to compare the results of different decision analysis methods. A Safe Route Planner requires a special evaluation method to compare the results and penalizes bad choices.

5. Research methodology

Fig. 2 depicts the overall research procedure to apply multi criteria decision analysis methods and evaluate the results. First, Safe Route Planner uses Dijkstra's algorithm (Dijkstra, 1959) and generates *M* alternative paths from a source to a given destination. Followings are the paths and their definitions:

Path A: Shortest path.
Path B: Fastest path.
Path C: Safest path.
Path D: A variant of the shortest/fastest path with elimination of the edge with the highest crash rate (alternative 4).
Path E: Another variant of the shortest/fastest path with elimination of the edge with the highest crash rate (alternative 5).
⋮
Path M: Mth variant of the shortest/fastest path.

In some cases, depending on the source and the destination, two or more alternative paths might be the same. For instance, if there is no accident on the fastest path, the same path is considered as the safest path. However, if the travel distance is long enough, these alternatives will be different from each other. Paths *D* to *M*, which are variants of the shortest path or the fastest path, are generated by removing a road segment with the highest accident rate on the shortest path or fastest path and rerunning Dijkstra's algorithm. If the result is different from existing alternative paths, it will be considered as a new alternative path. Although it is possible to add more alternative paths such as a path that avoids tolls or highways, for simplicity of the pilot study, only these alternative paths are considered.

Having multiple different paths from a source to a given destination with different characteristics, makes it a difficult decision for the users to select which one is the right choice. The decision-making system will assist the users to select the best alternative path that is the overall goal. The criteria that contribute to the overall goal are the safety level of the roads, the travel time, and the distance. Based on the availability of data, it is possible to add more criteria such as road shape and traffic volume.

For each alternative path, {*A, B, C, D, …, M*}, the total distance, the travel time and the total safety weight of the path must be calculated as shown in Eqs. (4), (5) and (6). The total safety weight presented in this research is based on the weighted crash rate and the distance to eliminate the effect of the driving speed; this is in contrast with the safest path calculation that is based on the weighted crash rate and the travel time.

$$\forall \text{path} \in \{\text{A}, .., \text{M}\}. \quad TravelTimeSeconds_{path} = \sum_{Src \xrightarrow{i,j \in Nodes} Dest} \frac{hDist(n_i, n_j) \times 3600}{Speed(n_i, n_j)} \tag{4}$$

$$\forall \text{path} \in \{\text{A}, .., \text{M}\}. \quad TotalDist_{path} = \sum_{Src \xrightarrow{i,j \in Nodes} Dest} hDist(n_i, n_j) \tag{5}$$

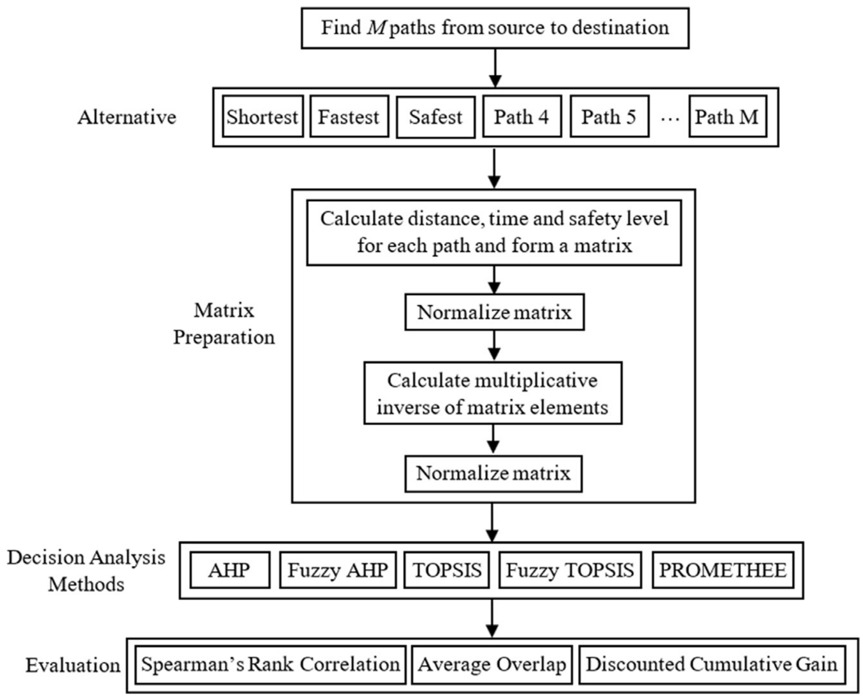

Fig. 2. Overall research procedure.

$$\forall \text{path} \in \{\text{A}, \ .., \text{M}\}.\ SafetyLevel_{path} = \sum_{Src \xrightarrow{i,j \in Nodes} Dest} WCR\left(n_i, n_j\right) \times hDist\left(n_i, n_j\right) \tag{6}$$

where *hDist* is the distance in kilometer between the two nodes that is calculated by the haversine formula; *3600* is the number of seconds in an hour; *n* is the node; *speed* is in km/hour, and *WCR* is the weighted crash rate between two nodes.

Matrix $A = [a_{ij}]_{m \times n}$ is constructed with a row for each alternative path, and columns representing safety level of the paths, travel times, and travel distances. The matrix preparation step transforms the matrix into a suitable format that can be used by the decision-making methods. Different columns have different scales; therefore, the matrix must be normalized to transform values into comparable values and to have the same range of values for each of the columns of the matrix. Normalization is performed by dividing the elements of each column by the sum of the column as shown in Eq. (7). This results in having values ranging between zero and one.

$$b_{ij} = \frac{a_{ij}}{\sum_{i=1}^{m} a_{ij}} \tag{7}$$

Lower numbers in the matrix mean results that are more favorable. For instance, the safety level is based on the number of accidents on the path; therefore, lower numbers indicate higher safety. Similarly, lower numbers for the travel time and the total distance mean results that are more favorable. For the rest of the calculation, these numbers must be changed in a way that the higher number indicates a better choice. To achieve this, after the normalization step, the multiplicative inverse of each element of the matrix is calculated as shown in Eq. (8).

$$c_{ij} = \frac{1}{b_{ij}} \tag{8}$$

The resulting matrix must be again normalized as shown in Eq. (9) to keep the values between zero and one. In this matrix, a higher number for safety indicates safer path, higher numbers for distance and time mean shorter distance and shorter travel time, respectively.

$$d_{ij} = \frac{c_{ij}}{\sum_{i=1}^{m} c_{ij}} \tag{9}$$

Five different decision analysis methods are used to find the most suitable path based on the user's judgement and rank the alternative paths. Fig. 3 illustrates the overall method employed in this research to derive the weights of criteria and the final ranks of alternatives. First, AHP and Fuzzy AHP are applied to calculate the weight of criteria based on the user's input. Next, these weights are fed to the decision-making methods to determine the ranking of the alternative paths. AHP is the only decision-making system added to the Safe Route Planner, and it is considered as the baseline for the comparison of different MCDA methods. Finally, using three different evaluation methods, namely, Spearman's rank correlation coefficient, average overlap, and discounted cumulative gain, the result of decision analysis methods are evaluated.

Fig. 4 illustrates two alternative paths, namely, shortest (A) and fastest (B), from a source to a destination. The resulting matrix is presented in Fig. 5, and the final normalized matrix is presented in Fig. 6. This simple example will be used in the following sections to demonstrate the calculation steps of each decision analysis method. In the results section, two real case studies along with their results and evaluations are presented.

6. Analytical hierarchy process

The first step in AHP is to form a hierarchy structure of the problem (Decompositions). The ultimate goal is at the top level of the hierarchy and criteria will be immediately below it. Criteria can be further divided into sub-criteria and finally the alternatives at the bottom level. Fig. 7 shows a three level hierarchy.

The next step is constructing pairwise comparison matrices (Prioritization) as shown in Fig. 8. The set of homogeneous elements in each level of the hierarchy is compared with itself with respect to elements in an upper level. The elements in the column on the left are compared against elements placed in the row on

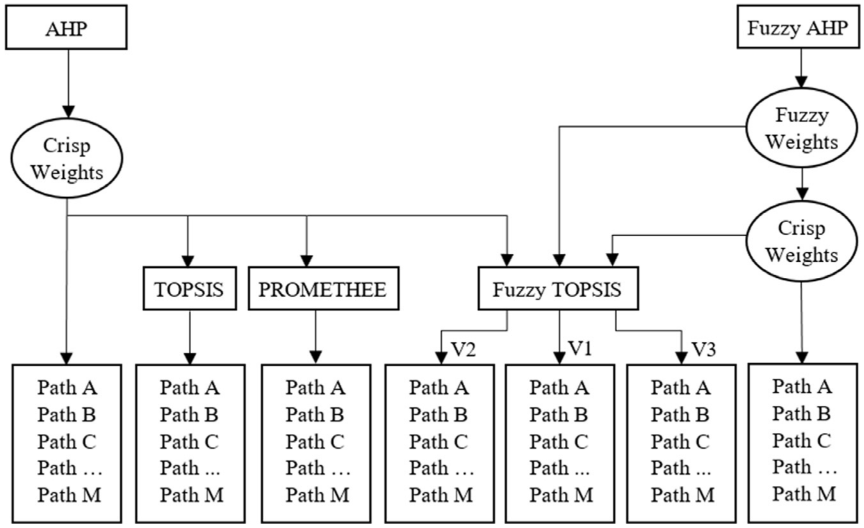

Fig. 3. Overall decision making systems employed.

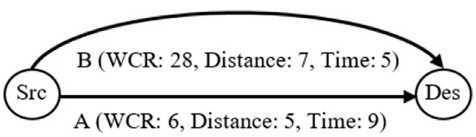

Fig. 4. A road network example.

$$\begin{array}{l} \\ \text{Shortest } (A) \\ \text{Fastest } (B) \end{array} \begin{array}{c} \begin{array}{ccc} \text{Safety} & \text{Distance} & \text{Time} \end{array} \\ \begin{bmatrix} 6 & 5 & 9 \\ 28 & 7 & 5 \end{bmatrix} \end{array}$$

Fig. 5. Two alternative paths and their properties.

$$\begin{array}{l} \\ \text{Shortest } (A) \\ \text{Fastest } (B) \end{array} \begin{array}{c} \begin{array}{ccc} \text{Safety} & \text{Distance} & \text{Time} \end{array} \\ \begin{bmatrix} 0.8235 & 0.5833 & 0.3571 \\ 0.1765 & 0.4167 & 0.6429 \end{bmatrix} \end{array}$$

Fig. 6. Final result of the sample network (higher numbers indicate more favorable results).

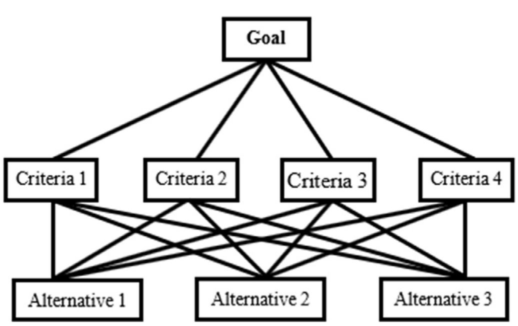

Fig. 7. Three level hierarchy.

top. A scale of numbers as shown in Table 2 is used for the comparison to assign numbers based on relative importance. The scale indicates how many times an element is more important over another element. Let *A* be an n-by-n matrix and $E_1, E_2, \ldots, E_n$ be the set of elements as shown in Fig. 8. For instance, 7 must be entered in a_{13} if E_1 is strongly more important than E_3.

$$A = \begin{array}{c} \\ E_1 \\ E_2 \\ E_3 \\ \vdots \\ E_n \end{array} \begin{array}{c} \begin{array}{ccccc} E_1 & E_2 & E_3 & \cdots & E_n \end{array} \\ \begin{bmatrix} 1 & a_{12} & 7 & \cdots & a_{1n} \\ 1/a_{12} & 1 & & \cdots & a_{2n} \\ 1/7 & & 1 & \cdots & \vdots \\ \vdots & & & \vdots & \vdots \\ 1/a_{1n} & & & & 1 \end{bmatrix} \end{array}$$

Fig. 8. Pairwise comparison matrix.

Table 2
AHP Comparison Scale.

Intensity of Importance	Definition
1	Equal importance
3	Moderate importance
5	Strong importance
7	Very strong or demonstrated importance
9	Extreme importance
2, 4, 6, 8	Intermediate values

The matrices are positive and reciprocal, $a_{ji} = 1/a_{ij}$; therefore, 1/7 must be entered in a_{31}. The number of judgments for a matrix of order n is $n\times(n - 1)/2$ since the lower triangle elements are the inverse of the elements in the upper triangle. The main diagonal elements of the matrix are all equal to one since an element is equally important when compared with itself.

The third step is calculating a vector of priorities by finding the eigenvector with the largest eigenvalue. This vector gives the relative importance of the various criteria being compared. This can be calculated by raising the matrix to a large power, then adding values in each row and normalizing to obtain the priority vector. This process should be repeated until the difference between the priority vector obtained at the kth power and at $(k+1)$st power is less than some predetermined small value.

The matrix A is said to be consistent if $a_{ij}\, a_{jk} = a_{ik}$ for all i, j and k. In a positive reciprocal matrix, if the maximum eigenvalue (λ_{max}) is equal to n, it is a consistent matrix. If λ_{max} exceeds n, the matrix is inconsistent. To calculate λ_{max}, the matrix of comparison must be multiplied by the vector of priorities obtaining a new vec-

Table 3
Average Random Consistency Index (R.I.).

N	1	2	3	4	5	6	7	8	9	10
Random Index	0	0	0.52	0.89	1.11	1.25	1.35	1.40	1.45	1.49

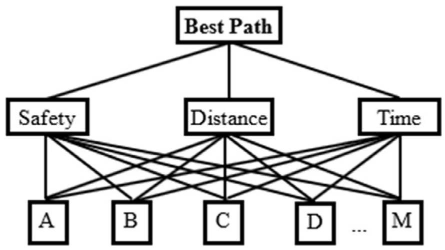

Fig. 9. AHP hierarchy structure for Safe Route Planner.

tor. Then, the first element of the new vector must be divided by the first element of the priority matrix, the second element of the new vector by the second element of the priority matrix and so on. The result will be another vector, and the maximum eigenvalue is the average of the elements of this vector.

To measure the inconsistency of a matrix (Consistency Index, CI), one should calculate $(\lambda_{max} - n)/(n - 1)$, and compare the result with the consistency index of a matrix of the same size with randomly chosen judgments that is called a Random Index (RI) as shown in Table 3. The consistency Ratio (CR) is the ratio of the consistency index to the average random index for the same order matrix. A CR of 0.10 or less is considered acceptable.

7. Deriving criteria weights with AHP

Fig. 9 illustrates the hierarchy structure of the Safe Route Planer problem with three criteria and *M* alternative paths. The next step is the judgement. The criteria are arranged into a square matrix, and the user makes a judgment about the relative importance of criteria against each other as shown in Fig. 10. As can be seen in Fig. 10, the safety level is favored strongly over the distance, as a result 5 is placed for safety against distance in the matrix in Fig. 11. Travel time is slightly favored over safety, thus 3 must be placed in row 3 column 1. Finally, in this example, the travel time is 7 times more important than the distance; therefore, 7 is entered in row 3 column 2.

After constructing the pairwise comparison matrix, the next step is to calculate the vector of priorities that shows the relative importance of the criteria. An easy method to calculate the vector of priorities if n<=3 is by normalizing the column (dividing the elements of each column by the sum of that column), and then calculating the average of each row. Fig. 12 shows the matrix of comparison after normalization.

The sum of the rows gives the column vector (0.8485, 0.2213, 1.9302). The average of the rows gives the vector of priorities as shown in Fig. 13. The priority vector indicates that the most influential factor for the user is the travel time (0.6434), followed by the safety level of the path (0.2828), and the least significant factor is the distance (0.0738).

At this point, the consistency of the matrix must be checked by calculating the maximum or principal eigenvalue ($\lambda_{\max}$) to make sure the user's judgement is consistent. The first step is multiplying the matrix of comparison by the vector of priorities to obtain a new vector as shown in Fig. 14.

The next step is to divide the first element of the new vector by the first element of the priority vector, the second element of the new vector by the second element of the priority vector, and the third element of the new vector by the third element of the priority vector. This calculation gives us a new vector that is (3.0629, 3.0121, 3.1215). Finally, to obtain $\lambda_{\max}$, the average of the elements in this vector is calculated which is 3.0655.

The next step is to calculate $(\lambda_{\max} - \mathrm{n})/(\mathrm{n} - 1)$ that is the consistency index, (3.0655 − 3) / (3 − 1) = 0.0327. Finally, the consistency ratio is calculated by dividing the consistency index by the random index for n = 3, 0.0327 / 0.58 = 0.0563. Since the consistency index is less than 0.10, it is acceptable and the judgement is considered consistent.

The final step in the process is to make a decision and find out which route is the best path for the driver. Fig. 15 illustrates the process of applying the vector of priorities to the matrix of alternatives.

Based on the results, the best path is alternative path *A*. Although based on the user's judgement the travel time is almost twice more important than the safety, alternative *A* with a longer travel time is selected since this path is almost five times safer than alternative *B*.

8. Deriving criteria weights with fuzzy AHP

In this section, Chang's (1996) extent analysis method on fuzzy AHP is applied to derive the weights and rank the alternatives. It is necessary to briefly explain the basic definitions and algebraic operations on fuzzy sets, then study the extent analysis method on fuzzy AHP. An object *x* is a member of classical set *A*, and it is represented by membership function $\mu A(x) = 1$ if it is a member of *A*, and $\mu A(x) = 0$ if object *x* is not a member of set *A*. In fuzzy set theory, the membership value ranges between 0 and 1, and it represents the degree in which an object *x* belongs to a fuzzy set. Different shapes of fuzzy membership functions exist, such as triangular, trapezoidal, linear, S-curve, or Gaussian. Triangular membership functions are the simplest membership model as they are defined by only three numbers (Pedrycz et al., 2011).

A Triangular fuzzy number is presented by the triplet, $M = (l,m,u)$, where *l* stands for lower value, *m* stands for modal value, *u* stands upper value of *m*, and $l \le m \le u$. If $l = m = u$, it is a nonfuzzy number. The membership function of the triangular fuzzy number is defined as in Eq. (10), and it is shown in Fig. 16.

$$\mu \mathrm{A}(\mathrm{x}) = \begin{cases} 0, \ if \ x \ \le l \\ \dfrac{x-l}{m-l}, \ l \ \le x \ \le m \\ \dfrac{u-x}{u-m}, \ m \ \le x \ \le u \\ 0, \ if \ x \ > u \end{cases} \tag{10}$$

Decision makers express their decisions regarding the importance of the criteria and alternatives in fuzzy numbers with triangular membership functions. There are many different triangular fuzzy scales to represent the judgement scale (Ishizaka). Chang's (1996) triangular fuzzy scale is presented in Table 4.

Consider $M_1 = (l_1, m_1, u_1)$ and $M_2 = (l_2, m_2, u_2)$ to be two triangular fuzzy numbers. The algebraic operations are as follows:

1. $\mathrm{M}_1 \oplus \mathrm{M}_2 = (\mathrm{l}_1+\mathrm{l}_2,\ \mathrm{m}_1+\mathrm{m}_2,\ \mathrm{u}_1+\mathrm{u}_2)$
2. $M_1 \odot M_2 \approx (l_1 l_2, m_1 m_2, u_1 u_2)$
3. $\lambda \odot M_2 \approx (\lambda l_1, \lambda m_1, \lambda u_1) \lambda\ \mathrm{M}_2 \approx (\lambda\ \mathrm{l}_1,\ \lambda\ \mathrm{m}_1,\ \lambda \mathrm{u}_1)$
4. $(\mathrm{l}_1,\ \mathrm{m}_1,\ \mathrm{u}_1)^{-1} \approx (1/\ \mathrm{u}_1,\ 1/\mathrm{m}_1,\ 1/\ \mathrm{l}_1)$

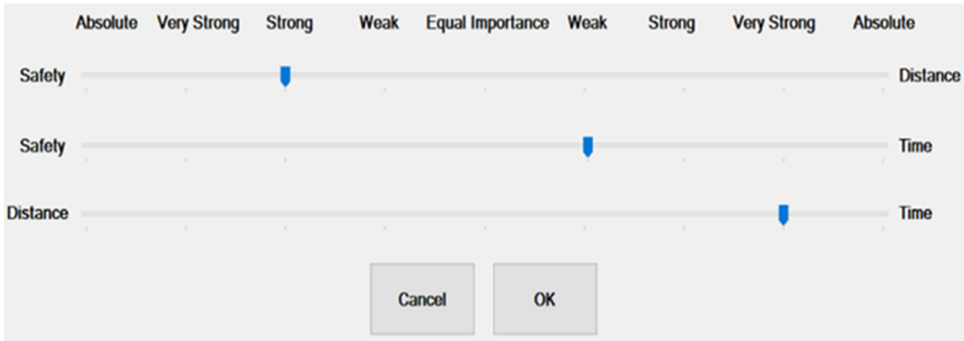

Fig. 10. Safe Route Planner's AHP input.

$$\begin{array}{l} \\ \text{Safety} \\ \text{Distance} \\ \text{Time} \end{array} \begin{array}{c} \begin{array}{ccc} \text{Safety} & \text{Distance} & \text{Time} \end{array} \\ \begin{bmatrix} 1 & 5 & 1/3 \\ 1/5 & 1 & 1/7 \\ 3 & 7 & 1 \end{bmatrix} \end{array} = \begin{array}{l} \\ \text{Safety} \\ \text{Distance} \\ \text{Time} \end{array} \begin{array}{c} \begin{array}{ccc} \text{Safety} & \text{Distance} & \text{Time} \end{array} \\ \begin{bmatrix} 1.0000 & 5.0000 & 0.3333 \\ 0.2000 & 1.0000 & 0.1429 \\ 3.0000 & 7.0000 & 1.0000 \end{bmatrix} \end{array}$$

Fig. 11. User's judgement matrices.

$$\begin{array}{l} \\ \text{Safety} \\ \text{Distance} \\ \text{Time} \end{array} \begin{array}{c} \begin{array}{ccc} \text{Safety} & \text{Distance} & \text{Time} \end{array} \\ \begin{bmatrix} 0.2381 & 0.3846 & 0.2258 \\ 0.0476 & 0.0769 & 0.0968 \\ 0.7143 & 0.5385 & 0.6774 \end{bmatrix} \end{array}$$

Fig. 12. Matrix of comparison.

$$\begin{array}{c} \begin{array}{ccc} \text{Safety} & \text{Distance} & \text{Time} \end{array} \\ \begin{bmatrix} 0.2828 & 0.0738 & 0.6434 \end{bmatrix} \end{array}$$

Fig. 13. Criteria weights obtained from AHP.

$$\begin{bmatrix} 1.0000 & 5.0000 & 0.3333 \\ 0.2000 & 1.0000 & 0.1429 \\ 3.0000 & 7.0000 & 1.0000 \end{bmatrix} \times \begin{bmatrix} 0.2828 \\ 0.0738 \\ 0.6434 \end{bmatrix} = \begin{bmatrix} 0.8662 \\ 0.2223 \\ 2.0084 \end{bmatrix}$$

Fig. 14. Intermediary vector to calculate consistency.

$$\begin{array}{l} \text{Path} \\ \text{A} \\ \text{B} \end{array} \begin{array}{c} \begin{array}{ccc} \text{Safety} & \text{Distance} & \text{Time} \end{array} \\ \begin{bmatrix} 0.8235 & 0.5833 & 0.3571 \\ 0.1765 & 0.4167 & 0.1537 \end{bmatrix} \end{array} \times \begin{array}{c} \text{Priorities} \\ \begin{bmatrix} 0.2828 \\ 0.0738 \\ 0.6434 \end{bmatrix} \end{array} = \begin{array}{c} \text{Results} \\ \begin{bmatrix} 0.5057 \\ 0.4943 \end{bmatrix} \end{array}$$

Fig. 15. Finding the best path by AHP.

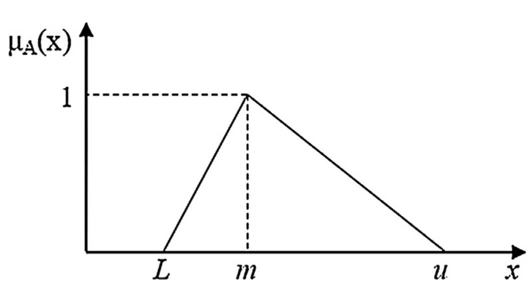

Fig. 16. Triangular fuzzy number.

Table 4
Chang's triangular fuzzy scale.

Linguistic terms	Saaty scale	Fuzzy Triangular Scale (*l*,*m*,*u*)
Equal importance	1	(1, 1, 1)
Moderate importance	3	(2/3, 1, 3/2)
Strong importance	5	(3/2, 2, 5/2)
Very strong importance	7	(5/2, 3, 7/2)
Extreme importance	9	(7/2, 4, 9/2)

$$\begin{array}{l} \\ \text{Safety} \\ \text{Distance} \\ \text{Time} \end{array} \begin{array}{c} \begin{array}{ccc} \text{Safety} & \text{Distance} & \text{Time} \end{array} \\ \begin{bmatrix} (1,1,1) & (1.5, 2, 2.5) & (0.666, 1, 1.5) \\ (0.4, 0.5, 0.666) & (1,1,1) & (0.2857, 0.3333, 0.4) \\ (0.666, 1, 1.5) & (2.5, 3, 3.5) & (1,1,1) \end{bmatrix} \end{array}$$

Fig. 17. Fuzzy matrix of comparison.

The first step of fuzzy AHP is to form the fuzzy pairwise comparison matrix using a fuzzy triangular scale, as shown in Fig. 17.

According to Chang's (1996) method of extent analysis, each object is taken and extent analysis for each goal, g_i, is performed. As a result, m extent analysis values for each object can be obtained by using the following signs:

$$M_{g_i}^1, M_{g_i}^2, \ \ldots, \ M_{g_i}^m, \ i = 1, \ 2, \ 3, \ \ldots n$$

where all the $M_{g_i}^j$ $(j = 1, \ 2, \ \ldots, \ m)$ are triangular fuzzy numbers.

The next step is to calculate the value of the fuzzy synthetic extent with respect to the *i*th criterion as shown in Eq. (11).

$$S_i = \sum_{j=1}^{m} M_{g_i}^j \odot \left[\sum_{i=1}^{n} \sum_{j=1}^{m} M_{g_i}^j \right]^{-1} \tag{11}$$

Fuzzy numbers must be compared to find the estimated vector of priorities. In order to find which fuzzy triangular number is the greatest among several fuzzy numbers, it is essential to evaluate the degree of possibility $M_1 \geq M_2$ where $M = (l,m,u)$ as shown in Eq. (12).

$$V(M_1 \geq M_2) \ = \ SUPREMUM_{x \ \geq y} \left[\min(\mu_{M_1}(x), \ \mu_{M_2}(y)) \right] \tag{12}$$

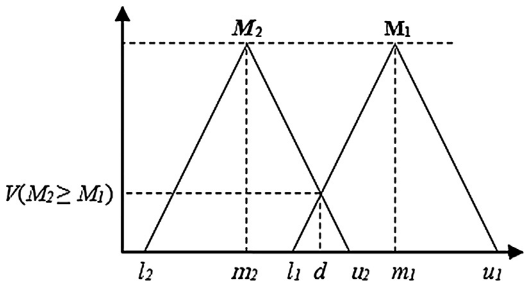

Fig. 18. The intersection between M_1 and M_2.

This can be equivalently expressed as Eq. (13).

$$V(M_2 \geq M_1) = \begin{cases} 1, & \text{iff } m_2 \geq m_1 \\ 0, & 0, \text{ if } l_1 \geq u_2 \\ \mu_{M_1}(d) = \dfrac{l_1 - u_2}{(m_2 - u_2) - (m_1 - l_1)} & \text{otherwise} \end{cases} \tag{13}$$

where d is the ordinate of the highest intersection between μ_{M1} and μ_{M2} as shown in Fig. 18.

If the two triangularvv fuzzy numbers do not intersect, $\mu(d)$ is zero (Zhu et al., 1999). To compare M_1 and M_2, both the values of $V(M_1 \geq M_2)$ and $V(M_2 \geq M_1)$ are required.

The degree of possibility for a convex fuzzy number to be greater than k convex fuzzy numbers M_i ($i = 1, 2 \ldots\ldots k$) can be defined by Eq. (14).

$$\begin{aligned} V(M \geq M_1, M_2, \ldots, M_k) \\ = V[(M \geq M_1)\text{and}(M \geq M_2) \text{ and } \ldots \text{ and } (M \geq M_k)] \\ = \min V(M \geq M_i) i = 1, 2, \ldots, k. \end{aligned} \tag{14}$$

Based on the previous calculation, the weight vector is computed as shown in Eq. (15).

$$W' = \left(d'(M_1),\ d'(M_2),\ \ldots,\ d'(M_n)\right)^T \tag{15}$$

where $d'(M_i)$ is calculated as shown in Eq. 16.

$$d'(M_i) = minV(S_i \geq S_k) k = 1, 2, \ldots, n \text{ and } k \neq i \tag{16}$$

By normalization, the non-fuzzy priority weight vector can be obtained as shown in Eq. (17):

$$W = (d(M_1), d(M_2),\ \ldots,\ d(M_n))^T \tag{17}$$

Based on the fuzzy AHP comparison matrix given in Fig. 17 the fuzzy weights (S_1, S_2, S_3) are obtained.

$$\begin{aligned} S_1 &= (3.1666,\ 4,\ 5) \odot \left(\frac{1}{13.0666}, \frac{1}{10.8333}, \frac{1}{9.0189}\right) \\ &= (0.2423,\ 0.3692,\ 0.5543) \\ S_2 &= (1.6857,\ 1.8333,\ 2.0666) \odot \left(\frac{1}{13.0666}, \frac{1}{10.8333}, \frac{1}{9.0189}\right) \\ &= (0.1290,\ 0.1692,\ 0.2291) \\ S_3 &= (4.1666, 5,\ 6) \odot \left(\frac{1}{13.0666}, \frac{1}{10.8333}, \frac{1}{9.0189}\right) \\ &= (0.3189,\ 0.4615,\ 0.6652) \end{aligned}$$

Based on Eq. (13), the degree of possibility between the criteria must be calculated. For instance, $V(S_1 \geq S_2)$ equals one, since the modal value of S_1 (0.3692) is greater than the modal value of S_2 (0.1692).

$V(S_1 \geq S_2) = 1$
$V(S_1 \geq S_3) = \frac{0.3189-0.5543}{(0.3692-0.5543)-(0.4615-0.3189)} = 0.7183$
$V(S_2 \geq S_1) = 0$ since $l_1 (0.2423) \geq u_2\ (0.2291)$
$V(S_2 \geq S_3) = 0$ since $l_1\ (0.3189) \geq u_2\ (0.2291)$

$$\begin{array}{cc} & \text{Safety Distance Time} \\ \text{Path} & \\ \text{A} & \\ \text{B} & \end{array} \begin{bmatrix} 0.8235 & 0.5833 & 0.3571 \\ 0.1765 & 0.4167 & 0.1537 \end{bmatrix} \times \overset{\text{Priorities}}{\begin{bmatrix} 0.4180 \\ 0 \\ 0.5820 \end{bmatrix}} = \overset{\text{Results}}{\begin{bmatrix} 0.5520 \\ 0.4480 \end{bmatrix}}$$

Fig. 19. Finding the best path by fuzzy AHP.

$V(S_3 \geq S_1) = 1$
$V(S_3 \geq S_2) = 1$
$d'(M_i)$ is calculated based on Eq. (16). C_1, C_2, and C_3 represent safety, distance and time criteria, respectively.
$d'(C_1) = V(S_1 > S_2, S_3) = \min(1, 0.7183) = 0.7183$
$d'(C_2) = V(S_2 > S_1, S_3) = \min(0, 0) = 0$
$d'(C_3) = V(S_3 > S_1, S_2) = \min(1, 1) = 1$

Using Eq. (15), the weight vector is obtained as follows:

$$W' = (0.7183, 0, 1)$$

By normalizing the weight, the normalized weight of criteria are obtained. Where W is a nonfuzzy weight.

$$W = (0.4180, 0, 0.5820)$$

The final step is applying the normalized weight of criteria to the matrix of alternatives as shown in Fig. 19.

Based on the results in Fig. 19, the order of selected alternative paths are {A, B}.

9. TOPSIS

The TOPSIS procedure is based on the following steps:

Step 1. Given matrix $A(x)_{mn}$ Calculate the normalized decision matrix as shown in Eq. (18).

$$r_{ij}(x) = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{m} x_{ij}^2}},\ i = 1,\ \ldots,\ m;\ j = 1,\ \ldots,\ n. \tag{18}$$

Step 2. Based on the weights of criteria, calculate the weighted normalized decision matrix as shown in Eq. (19).

$$v_{ij} = w_j r_{ij}, i = 1, \ldots, m; j = 1, \ldots, n. \tag{19}$$

where w_j is the weight of the jth criterion.

Step 3. Determine the positive ideal and negative ideal solutions as shown in Eq. (20).

$$\begin{aligned} A^+ &= \left\{v_1^+ \cdot v_2^+ \cdot \ldots, v_n{}^+\right\} \text{ where } v_j^+ = \left\{\max\left(v_{ij}\right) \text{ if } j \in B; \min\left(v_{ij}\right) \text{ if } j \in C\right\} \\ A^- - &= \left\{v_1^- \cdot v_2^- \cdot \ldots, v_n^-\right\} \text{ where } v_j^- = \left\{\min\left(v_{ij}\right) \text{ if } j \in B; \max\left(v_{ij}\right) \text{ } if\ j \in C\right\} \end{aligned} \tag{20}$$

where B is associated with the benefit and C is associated with the cost criteria.

Step 4. Based on Eq. (21), calculate the separation from positive ideal and negative ideal solutions for each alternative.

$$\begin{aligned} S_i^+ &= \sqrt{\sum_{j=1}^{n} \left(v_{ij} - v_j^+\right)^2},\ i = 1,\ \ldots,\ m \\ S_i^- &= \sqrt{\sum_{j=1}^{n} \left(v_{ij} - v_j^-\right)^2},\ i = 1,\ \ldots,\ m \end{aligned} \tag{21}$$

Step 5. Calculate the relative closeness to the ideal solution as shown in Eq. (22).

$$C_i^- = \frac{S_i^-}{S_i^+ + S_i^-},\ i = 1,\ \ldots,\ m \tag{22}$$

$$\begin{array}{cc} & \begin{array}{ccc} \text{Safety} & \text{Distance} & \text{Time} \end{array} \\ \text{Path} & \\ r_{ij}(x) = \begin{array}{c} A \\ B \end{array} & \begin{bmatrix} 0.2095 & 0.5812 & 0.8742 \\ 0.9778 & 0.8137 & 0.4856 \end{bmatrix} \end{array}$$

Fig. 20. Normalized alternative paths.

$$\begin{array}{cc} & \begin{array}{ccc} \text{Safety} & \text{Distance} & \text{Time} \end{array} \\ \text{Path} & \\ v_{ij}(x) = \begin{array}{c} A \\ B \end{array} & \begin{bmatrix} 0.0592 & 0.0429 & 0.5625 \\ 0.2765 & 0.0601 & 0.3124 \end{bmatrix} \end{array}$$

Fig. 21. Weighted normalized decision matrix.

Table 5
Weights of criteria in fuzzy TOPSIS.

Linguistic terms	Fuzzy triangular numbers
Very Low (VL)	(0, 0, 0.1)
Low (L)	(0, 0.1, 0.3)
Medium Low (ML)	(0.1, 0.3, 0.5)
Medium (M)	(0.3, 0.5, 0.7)
Medium High (MH)	(0.5, 0.7, 0.9)
High (H)	(0.7, 0.9, 1.0)
Very High (VH)	(0.9, 1.0, 1.0)

Finally, rank the alternatives based on C_i^-.

In order to apply TOPSIS to the sample example, the matrix that is given in Fig. 5 is used. Next, the matrix is normalized based on Step 1 as shown in Fig. 20.

Based on Step 2, calculate the weighted normalized decision matrix. The weight of criteria were obtained in AHP calculation as (0.2828, 0.0738, 0.6434). Fig. 21 illustrates the weighted normalized decision matrix.

The positive and negative ideal solutions are obtained based on Step 3:

$$A^+ = \{0.0592, 0.0429, 0.3124\}$$
$$A^- - = \{0.2765, 0.0601, 0.5625\}$$

Step 4. The separation from positive ideal and negative ideal solutions is calculated for each alternative.

$$S_i^+ = \{0.2502, 0.2179\}$$
$$S_i^- = \{0.2179, 0.2502\}$$

Step 5. Calculate the relative closeness to the ideal solution.

$$C_i^- = \{0.4655\ (A), 0.5345\ (B)\}$$

If there are more than two alternative paths available, the resulting C_i^- must be normalized. Based on the results, the order of selected alternative paths are $\{B, A\}$.

10. Fuzzy TOPSIS

The steps of Chen's (2000) fuzzy TOPSIS are as follows.

Step 1. The weight of criteria must be expressed in linguistic terms as shown in Table 5.

Step 2. The linguistic ratings presented in Table 6 are adopted to rate the alternatives with respect to each criterion.

Step 3. The linguistic terms are converted into triangular fuzzy numbers to construct the fuzzy decision matrix and the fuzzy weight of the each criterion.

Step 4. Construct normalized fuzzy decision matrix R as shown in Eq. (23).

$$R = \left[r_{ij}\right]_{m\times n}$$

Table 6
Linguistic ratings and associated fuzzy triangular numbers in fuzzy TOPSIS.

Linguistic terms	Fuzzy triangular numbers
Very Poor (VP)	(0, 0, 1)
Poor (P)	(0, 1, 3)
Medium Poor (MP)	(1, 3, 5)
Fair (F)	(3, 5, 7)
Medium Good (MG)	(5, 7, 9)
Good (G)	(7, 9, 10)
Very Good (VG)	(9, 10, 10)

$$\begin{aligned} r_{ij} &= \left(\frac{a_{ij}}{c_j^*}, \frac{b_{ij}}{c_j^*}, \frac{c_{ij}}{c_j^*}\right), \ j \in B; \\ r_{ij} &= \left(\frac{a_j^-}{c_{ij}}, \frac{a_j^-}{b_{ij}}, \frac{a_j^-}{a_{ij}}\right), \ j \in C; \\ c_j^* &= {}_i^{max} c_{ij} \ \text{ if } j \in B \\ a_j^- &= {}_i^{min} a_{ij} \ \text{ if } j \in C \end{aligned} \tag{23}$$

where B is associated with the benefit, and C is associated with the cost criteria.

Step 5. Construct the weighted normalized fuzzy decision matrix V as shown in Eq. (24).

$$\begin{aligned} V &= \left[v_{ij}\right]_{m\times n} \\ v_{ij} &= r_{ij} \odot w_j \end{aligned} \tag{24}$$

Step 6. Construct fuzzy positive-ideal solution (A^*) and fuzzy negative-ideal solution (A^-) as shown in Eqs. (25).

$$\begin{aligned} A^* &= (v_1^*, \ v_2^*, \ \ldots, \ v_n^*) \\ A^- &= \left(v_1^-, \ v_2^-, \ \ldots, \ v_n^-\right) \end{aligned} \tag{25}$$

where $v_j^* = (1, 1, 1)$ and $v_j^- = (0, 0, 0)$, j = 1, 2, ..., n.

Step 7. Calculate the distance of each alternative from A^* and A^- using Eqs. (26).

$$\begin{aligned} d_i^* &= \sum_{j=1}^{n} d\left(v_{ij}, \ v_j^*\right), \ i = 1, \ 2, \ \ldots, \ m. \\ d_i^- &= \sum_{j=1}^{n} d\left(v_{ij}, \ v_j^-\right), \ i = 1, \ 2, \ \ldots, \ m. \end{aligned} \tag{26}$$

where $d(m, n) = \sqrt{\frac{1}{3}\left[(m_1 - n_1)^2 + \ (m_2 - n_2)^2 + \ (m_3 - n_3)^2\right]}$

Step 8. Based on Eq. (27), calculate closeness coefficient to determine the ranking order of alternatives.

$$CC_i = \frac{d_i^-}{d_i^* + d_i^-}, \ i \ = \ 1, \ 2, \ \ldots, \ m. \tag{27}$$

Step 9. Determine the ranking of alternatives according to the closeness coefficient.

Sample calculation of Fuzzy TOPSIS:

For the calculation of Fuzzy TOPSIS, three methods are applied with different weights obtained from fuzzy AHP and traditional AHP.

Method 1 (Fuzzy TOPSIS v1): This method is based on the fuzzy weights obtained from Fuzzy AHP, and the sample alternative path matrix. The alternative paths are illustrated in Fig. 22.

Based on the linguistic terms, rate the alternatives with respect to each criterion as shown in Fig. 23.

$$\begin{array}{c c} & \begin{array}{ccc} \text{Safety} & \text{Distance} & \text{Time} \end{array} \\ \begin{array}{c} \text{Shortest } (A) \\ \text{Fastest } (B) \end{array} & \begin{bmatrix} 0.8235 & 0.5833 & 0.3571 \\ 0.1765 & 0.4167 & 0.6429 \end{bmatrix} \end{array}$$

Fig. 22. Alternative paths and their properties.

$$\begin{array}{c c} & \begin{array}{ccc} \text{Safety} & \text{Distance} & \text{Time} \end{array} \\ \begin{array}{c} \text{Shortest } (A) \\ \text{Fastest } (B) \end{array} & \begin{bmatrix} \text{VG} & \text{VG} & \text{VP} \\ \text{VP} & \text{VP} & \text{VG} \end{bmatrix} \end{array}$$

Fig. 23. Alternative paths and their properties in linguistic terms.

$$\begin{array}{c c} & \begin{array}{ccc} \text{Safety} & \text{Distance} & \text{Time} \end{array} \\ \begin{array}{c} \text{Shortest } (A) \\ \text{Fastest } (B) \end{array} & \begin{bmatrix} (9, 10, 10) & (9, 10, 10) & (0, 0, 1) \\ (0, 0, 1) & (0, 0, 1) & (9, 10, 10) \end{bmatrix} \end{array}$$

Fig. 24. Alternative paths and their properties in triangular fuzzy numbers.

$$\begin{array}{c c} & \begin{array}{ccc} \text{Safety} & \text{Distance} & \text{Time} \end{array} \\ \begin{array}{c} \text{Shortest } (A) \\ \text{Fastest } (B) \end{array} & \begin{bmatrix} (0.9, 1, 1) & (0.9, 1, 1) & (0, 0, 0.1) \\ (0, 0, 0.1) & (0, 0, 0.1) & (0.9, 1, 1) \end{bmatrix} \end{array}$$

Fig. 25. Alternative paths and their normalized properties in triangular fuzzy numbers.

Table 7
Distance of alternatives from fuzzy positive and negative ideal solutions.

Path	A^*	A^-
A	2.4427	0.6206
B	2.5252	0.5411

Construct the fuzzy decision matrix by converting linguistic terms into the associated triangular fuzzy numbers as shown in Fig. 24.

Construct the normalized fuzzy decision matrix as shown in Fig. 25.

Constructing the weighted normalized fuzzy decision matrix based on the triangular weights obtained from fuzzy AHP which are ((0.2423, 0.3692, 0.5543), (0.1290, 0.1692, 0.2291), (0.3189, 0.4615, 0.6652)). Fig. 26 illustrates the weighted normalized fuzzy decision matrix.

Calculate the distance of each Alternative from fuzzy positive-ideal solution (A^*) and fuzzy negative-ideal solution (A^-) as shown in Table 7.

Calculate the closeness coefficient.

$CC_1 = 0.2026, CC_2 = 0.1765.$

The normalized weights of the paths are (0.5344, 0.4656). Based on the results, the order of selected alternative paths are {*A, B*}.

Method 2 (Fuzzy TOPSIS v2): The Crisp weights (0.2828, 0.0738, 0.6434) obtained from AHP are adopted. The weights are fuzzified using linguistic terms and then the triangular fuzzy weights are obtained as (0, 0.1, 0.3), (0, 0, 0.1), (0.5, 0.7, 0.9). The calculated closeness coefficient based on these weights are as follows.

$CC_1 = 0.0941, CC_2 = 0.237.$

The normalized weights of the paths are (0.2842, 0.7158). Based on the results, the order of selected alternative paths are {*B, A*}.

Method 3 (Fuzzy TOPSIS v3): The crisp weights (0.4180, 0, 0.5820) obtained from Fuzzy AHP are adopted. The weights are fuzzified using linguistic terms and then the triangular fuzzy weights are obtained as (0.1, 0.3, 0.5), (0, 0, 0.1), (0.5, 0.7, 0.9). The calculated closeness coefficient based on these weights are as follows.

$CC_1 = 0.1447, CC_2 = 0.2403.$

The normalized weights of the paths are (0.3758, 0.6242). Based on the results, the order of selected alternative paths are {*B, A*}.

11. PROMETHEE

In this section, PROMETHEE II is applied to the sample road network that is given in section V. PROMETHEE II calculates a net flow for each alternative which represents the degree of dominance of alternatives. The ranking of the alternatives are determined based on the net flow. The net flow is calculated based on positive outranking flow and negative flow. The positive outranking flow shows how an alternative outranks all other alternatives, while the negative flow represents how an alternative is outranked by other alternatives.

Let $A = \{a_1, a_2, \ldots, a_n\}$ be a finite set of alternatives, and $F = \{g_1, g_2, \ldots, g_m\}$ be a set of evaluation criteria. The criteria can be either maximized or minimized. PROMETHEE requires the weights of relative importance of the criteria to be known. Let $W = \{w_1, w_2, \ldots, w_m\}$ be the weight of the criteria. The steps of PROMETHEE II are as follows:

Step 1. Calculate the unicriterion preference degrees

The preference structure of PROMETHEE is based on pairwise comparison. For each criterion, the deviation (d_j) between the evaluations of any two alternatives must be computed as shown in Eq. (28).

$$d_j = g_j(a) - g_j(b) \text{ where } (a, b) \in A,\ j = 1, 2, \ldots m \tag{28}$$

Based on the deviations and a predetermined preference function P_j, the preference degrees of alternatives are determined. The preference function $P_j(a,b)$ shows the extent of which the decision maker prefers alternative *a* over alternative *b* on criterion *j*. The preference degree ranges between zero and one. Zero means indifference between the alternatives and one means a total preference for one alternative over another. Six types of preference functions are proposed, namely usual criterion, U-shape criterion, V-shape criterion, level criterion, V-shape with indifference criterion, and Gaussian criterion (Brans & Mareschal 2005; Brans, Vincke, & Mareschal, 1986). In this research, the V-shape function is applied as shown in Eq. (29) and Fig. 27.

$$P(d) = \begin{cases} 0 & d \le 0 \\ d & 0 < d \le 1 \\ 1 & d > 1 \end{cases} \tag{29}$$

Step 2. Compute the global preference degrees

For each pair of alternatives, the global preference degree (π) over all the criteria must be computed. The overall preference degree of alternative *a* over alternative *b* is computed based on Eq. (30).

$$\pi(a, b) = \sum_{j=1}^{m} P_j(a, b) w_j \tag{30}$$

π (a, b) expresses the intensity of preference of alternative a over alternative b over all the criteria and π(b, a) indicates how b is preferred to a.

$$\begin{array}{l} \\ \text{Shortest }(A) \\ \text{Fastest }(B) \end{array} \begin{array}{c} \begin{array}{ccc} \text{Safety} & \text{Distance} & \text{Time} \end{array} \\ \begin{bmatrix} (0.2181, 0.3692, 0.5543) & (0.1161, 0.1692, 0.2291) & (0, 0, 0.0665) \\ (0, 0, 0.0554) & (0, 0, 0.0229) & (0.287, 0.4615, 0.6652) \end{bmatrix} \end{array}$$

Fig. 26. Alternative paths and their weighted normalized properties in triangular fuzzy numbers.

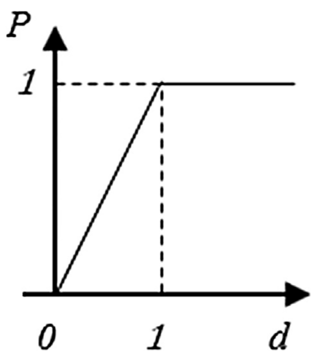

Fig. 27. V-shape function.

π(a, a) = 0
π(a, b) ≈ 0 indicates a weak global preference of a over b.
π(a, b) ≈ 1 indicates a strong global preference of a over b.

Step 3. Compute outranking flows.

For each alternative, the positive (ϕ^{+}), negative (ϕ^{-}) and net flows (ϕ) must be calculated as Eqs. (31), (32) and (33), respectively.

$$\phi^{+}(a) = \frac{1}{n-1} \sum_{x \in A} \pi(\mathrm{a, x}) \tag{31}$$

$$\phi^{-}(a) = \frac{1}{n-1} \sum_{x \in A} \pi(\mathrm{x, a}) \tag{32}$$

$$\phi(a) = \phi^{+}(a) - \phi^{-}(a) \tag{33}$$

The positive flow indicates how an alternative a is outranking all other alternatives, while the negative flow indicates how an alternative a is outranked by all the other alternatives. The positive flow is the power of an alternative, the higher the power the better the alternative. The negative flow is the weakness of an alternative, the lower the negative flow the better the alternative. The negative and positive flows range between zero and one, while the net flow ranges between minus one and one.

The range of net flow is confusing for the decision makers; therefore, in this research the range should be changed to [0, 1] interval and normalized as shown in Eq. (34).

$$\phi^{\sim}(a) = \frac{\phi(a) + 1}{\sum_{i=1}^{n} (\phi(a_i) + 1)} \tag{34}$$

Step 4. Based on the computed *$\phi^{\sim}(a)$ rank the alternatives.*

An alternative a outranks an alternative b if $\phi^{\sim}(a) > \phi^{\sim}(b)$
An alternative a is outranked by an alternative b if $\phi^{\sim}(a) < \phi^{\sim}(b)$
Two alternatives a and b are equivalent if $\phi^{\sim}(a) = \phi^{\sim}(b)$

Sample calculation of PROMETHEE:

Step 1. Compute the deviation (d_j) between the evaluations of any two alternatives as shown in Tables 8–10.

For instance, for safety criterion the deviation between alternatives A and B is 0.8235–0.1765 = 0.647. Similarly, the deviation between alternatives B and A is 0.1765–0.8235 = −0.647.

Steps 2 **and** 3**.** Compute the global preference degrees and the outranking flows as shown in Table 11.

Using the weight of criteria that are obtained in AHP calculation as (0.2828, 0.0738, 0.6434), the global preference degree of alternative A over alternative B is computed as follows:

$$\pi(A, B) = 0.647 \times 0.2828 + 0.1666 \times 0.0738 + 0 \times 0.6434 = 0.1953$$

The outranking flows calculation for alternative A is as follows:

$$\phi^{+}(A) = \frac{0 + 0.1953}{2} = 0.0976$$

$$\phi^{-}(A) = \frac{0 + 0.1839}{2} = 0.0919$$

$$\phi(A) = 0.0976 - 0.0919 = 0.0057$$

$$\phi^{\sim}(A) = \frac{(0.0057 + 1)}{(0.0057 + 1) + (-0.0057 + 1)} = 0.5028$$

Based on the results, the order of selected alternative paths are {A, B}.

12. Evaluation methodology

Once the decision analysis methods created rank lists of alternatives, it is necessary to quantify the similarity between the ranked lists. In this section, three different evaluation methods that will be used in the result section, namely Spearman's rank correlation coefficient, Average Overlap (AO), and Discounted Cumulative Gain (DCG) are explained. Spearman's rank correlation coefficient and Kendall's tau coefficient (Kendall, 1938) are two commonly used methods to compare ranked lists. Spearsman's rank correlation coefficient is defined as in Eq. (35).

$$R = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n\left(n^2 - 1\right)} \tag{35}$$

where d_i is the rank difference at position i, and n is the number of ranks.

The similarity measurement algorithms need to consider that the top of the list is more important than the bottom; therefore, a change in the ranks at the top of the list must be given more weight than a change at the bottom of the list. Average overlap discussed by Webber, Moffat, and Zobel (2010) is one approach to assign more weights to the top of the list. It is based on simple set overlap, where the user compares the overlap of the two rankings at incrementally increasing depths. Average overlap is shown in Eq. (36).

$$AO(S, T, K) = \frac{1}{k} \sum_{d=1}^{k} \frac{|S_d \cap T_d|}{d} \tag{36}$$

where S and T are ranking lists, K is the evaluation depth and d is depth.

Table 15 illustrates a sample calculation of the average overlap between two lists S = *{ABCDE}* and T = *{ABCED}*.

Table 8
Deviation between alternatives based on safety criterion.

Safety	Deviation (Safety Criterion)		Preference Degree (V-Shape Function)	
Alternatives	A	B	A	B
A	0	0.647	0	0.647
B	-0.647	0	0	0.1948

Table 9
Deviation between alternatives based on distance criterion.

Distance	Deviation (Distance Criterion)		Preference Degree (V-Shape Function)	
Alternatives	A	B	A	B
A	0	0.1666	0	0.1666
B	-0.1666	0	0	0

Table 10
Deviation between alternatives based on time criterion.

Time	Deviation (Time Criterion)		Preference Degree (V-Shape Function)	
Alternatives	A	B	A	B
A	0	-0.2858	0	0
B	0.2858	0	0.2858	0

Table 11
Global preference degrees and outranking flows.

	Global Preference Degrees (π)			Outranking Flows		
Alternatives	A	B	ϕ^+	ϕ^-	ϕ	ϕ^-
A	0.0000	0.1953	0.0976	0.0919	0.0057	0.5028
B	0.1839	0	0.0919	0.0976	-0.0057	0.4972

The average overlap ranges from zero, indicating disjoint (no similar items), to one indicating identical results. Since the lists of rankings in this research consist of the same items, the average overlap cannot be zero. The most different ranking between two lists results in an average overlap of 0.416, for instance if elements of the second list are in reverse order.

Discounted Cumulative Gain (DCG) proposed by Jarvelin and Kekalainen (2000, 2002) is widely used in information retrieval to evaluate the relevance of the results returned from different search engines. It computes the gain of results based on their positions. The first step to compute DCG is to have a relevance score for each of the items, for instance relevance scores can be {3, 2, 1, 0}. The highest score is assigned to the most relevant item, and the lowest score is assigned to a not relevant item. The ranked list must be turned into gained value list (G) by replacing the items by their relevance scores.

The cumulative gain (CG) can be computed by summing the relevance scores. Computing the discounted cumulative gain (DCG) requires a logarithmic discount function to reduce the gain score of alternatives continuously and gradually as their ranks increase. Discounted cumulative gain is presented in Eq. (37). Changing the base of the logarithm results in sharper or smoother discount. In this research, the base of the logarithm is selected to be three.

$$DCG = \sum_{i=1}^{n} \frac{G[i]}{\log_b(i+1)} \tag{37}$$

The normalized discounted cumulative gain (nDCG) can be computed as in Eq. (38).

$$nDCG = \frac{DCG}{IDCG} \tag{38}$$

where IDCG is the DCG of the ideal ranked list.

13. Results and comparison

In this section, the results of the different multi-criteria decision analysis methods applied to the Safe Route Planner are evaluated using two real case studies. The criteria weights of (0.2828, 0.0738, 0.6434) which were obtained using AHP are used to obtain the ranks. Based on the user's input and the AHP weight calculation presented in section VI, travel time was the most important factor for the user, followed by the safety level of the road. The travel distance was the least influential factor for the user.

The first evaluation of the Safe Route Planner using a decision-making system is based on a use case where the user intends to travel from the intersection of Forest Glen Road and MD-97, Silver Spring in Maryland to 1 Pooks Hill Road, Bethesda in Maryland. As shown in Fig. 28, Google Maps suggests three alternative paths, and the user must make an uninformed decision to select one. The first suggested path by Google Maps is a very risky path with a very high crash rate.

Using the Safe Route Planner, the users have the option to select the fastest, shortest, or safest path. In addition, the Analytical Hierarchy Process method assists drivers in selecting the best path. Figs. 29 and 30 demonstrate the alternative paths from the source to the destination using the Safe Route Planner. Road segments highlighted with blue color are considered safe, meaning that either there were no accidents on those road segments or there were a few accidents that can be ignored due to high traffic volume. Yellow indicates a medium level of safety, and red represents a high crash rate and high risk of an accident.

The matrix shown in Fig. 31 presents the alternative paths, the safety level, the distance and the travel time for each alternative path. The matrix clearly illustrates that the fastest path is too risky. The safety level of the fastest path is 115.9982, while the safety level of the safest path is 16.0389. The safest path is a relative term to indicate both the safest and the fastest combined. For instance, the shortest path in this evaluation is safer than the safest path. However, the travel time for shortest path is greater than the safest path, as a result the Safe Route Planner considers alternative C as the safest path.

Fig. 32 illustrates the matrix that is first normalized, then multiplicative inverse of the elements are calculated, and finally it is

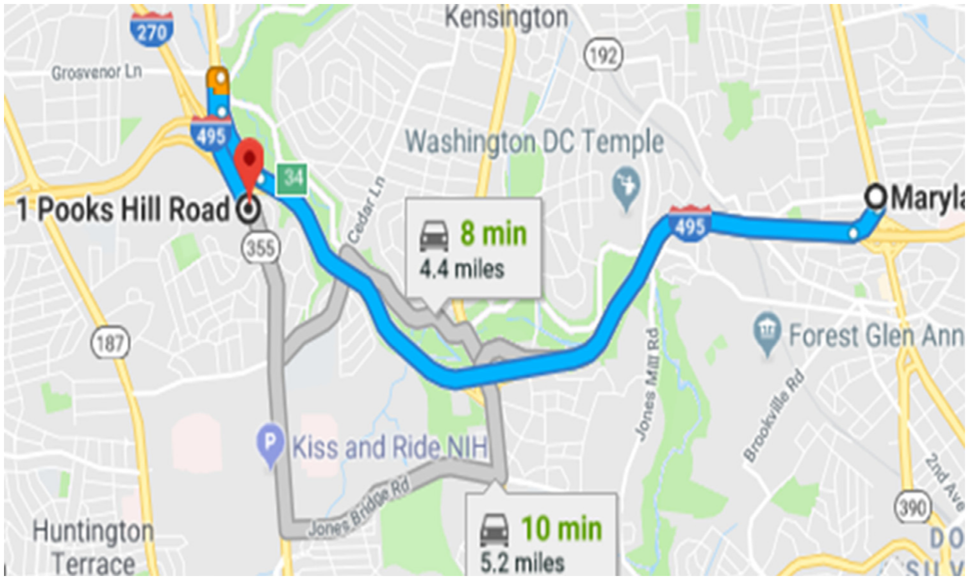

Fig. 28. Google Maps' suggested routes from MD-97 to 1 Pooks Hill Road.

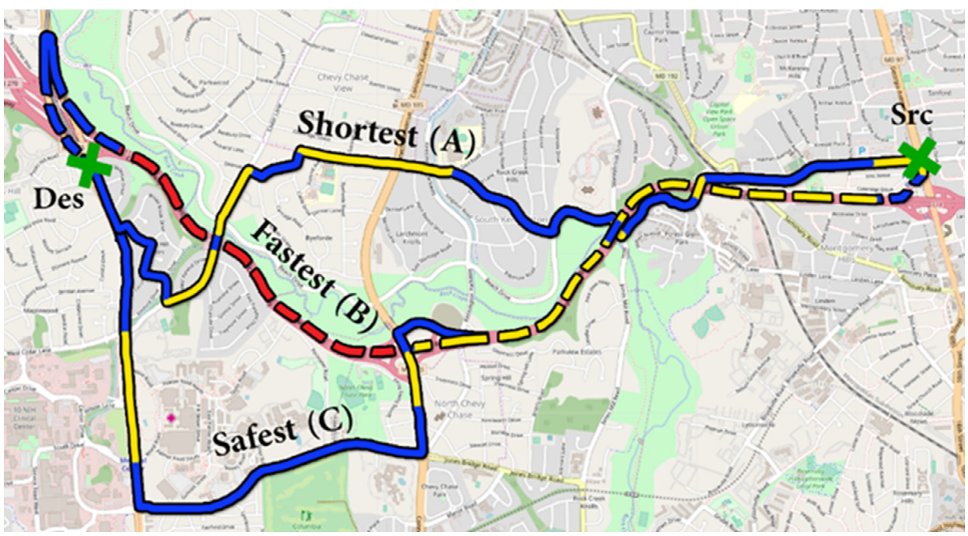

Fig. 29. Shortest, fastest and safest paths from MD-97 (Src) to 1 Pooks Hill Road (Des).

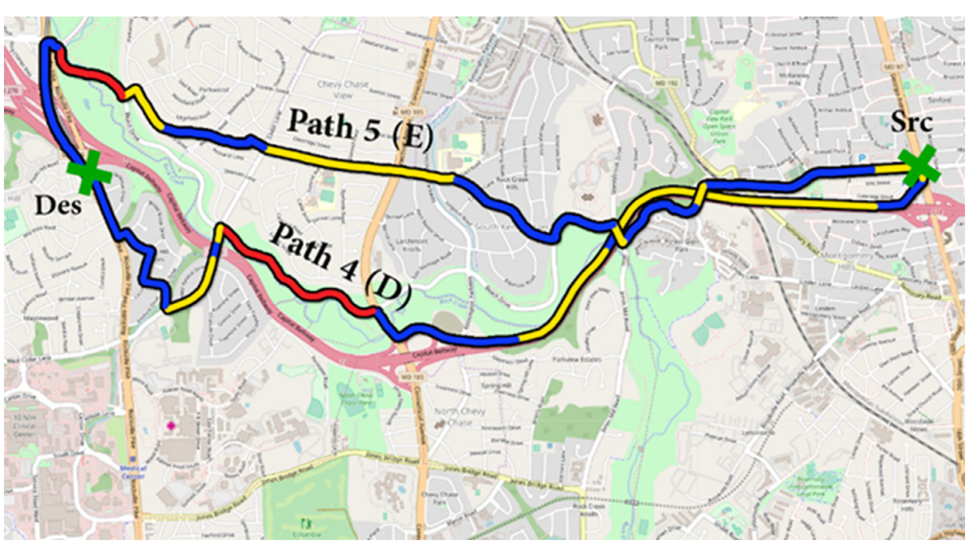

Fig. 30. Shortest path variants 1 and 2 from MD-97 (Src) to 1 Pooks Hill Road (Des).

Table 12
Comparison of alternative path weights assigned by different methods.

Path	AHP	Fuzzy AHP	TOPSIS	Fuzzy TOPSIS V1	Fuzzy TOPSIS V2	Fuzzy TOPSIS V3	PROMETHEE
Shortest (A)	0.1994	0.2119	0.1962	0.1832	0.1092	0.1376	0.1999
Fastest (B)	0.1891	0.1644	0.1128	0.1766	0.2828	0.2349	0.1973
Safest (C)	0.2144	**0.2255**	0.2372	0.1831	0.1768	0.1925	0.2036
Path 4 (D)	**0.2173**	0.2121	**0.2691**	**0.2834**	**0.3227**	**0.3009**	**0.2043**
Path 5 (E)	0.1796	0.1863	0.1848	0.1736	0.1085	0.1342	0.1949

Table 13
Comparison of alternative path ranks assigned by different methods.

Path	AHP	Fuzzy AHP	TOPSIS	Fuzzy TOPSIS V1	Fuzzy TOPSIS V2	Fuzzy TOPSIS V3	PROMETHEE
Shortest (A)	3	3	3	2	4	4	3
Fastest (B)	4	5	5	4	2	2	4
Safest (C)	2	**1**	2	3	3	3	2
Path 4 (D)	**1**	2	**1**	**1**	**1**	**1**	**1**
Path 5 (E)	5	4	4	5	5	5	5

Table 14
Spearsman's rank correlation coefficient between AHP and the other methods.

	Fuzzy AHP	TOPSIS	Fuzzy TOPSIS v1	Fuzzy TOPSIS v2	Fuzzy TOPSIS v3	PROMETHEE
AHP	0.8	0.9	0.9	0.7	0.7	1

$$
\begin{array}{l}
\\ \text{Shortest } (A) \\ \text{Fastest } (B) \\ \text{Safest } (C) \\ \text{Path 4 } (D) \\ \text{Path 5 } (E)
\end{array}
\begin{array}{c}
\begin{array}{ccc} \text{Safety} & \text{Distance} & \text{Time} \end{array} \\
\begin{bmatrix}
15.1057 & 6.7356 & 508.0000 \\
115.9982 & 7.6063 & 320.0000 \\
16.0389 & 8.2911 & 417.0000 \\
23.2725 & 6.7786 & 352.0000 \\
18.4128 & 6.9967 & 533.0000
\end{bmatrix}
\end{array}
$$

Fig. 31. Five alternative paths and their properties.

$$
\begin{array}{l}
\\ \text{Path A} \\ \text{Path B} \\ \text{Path C} \\ \text{Path D} \\ \text{Path E}
\end{array}
\begin{array}{c}
\begin{array}{ccc} \text{Safety} & \text{Distance} & \text{Time} \end{array} \\
\begin{bmatrix}
0.2824 & 0.2149 & 0.1612 \\
0.0368 & 0.1903 & 0.2560 \\
0.2659 & 0.1745 & 0.1964 \\
0.1833 & 0.2135 & 0.2327 \\
0.2316 & 0.2068 & 0.1537
\end{bmatrix}
\end{array}
$$

Fig. 32. Final results for Alternative paths (higher numbers indicate more favorable results).

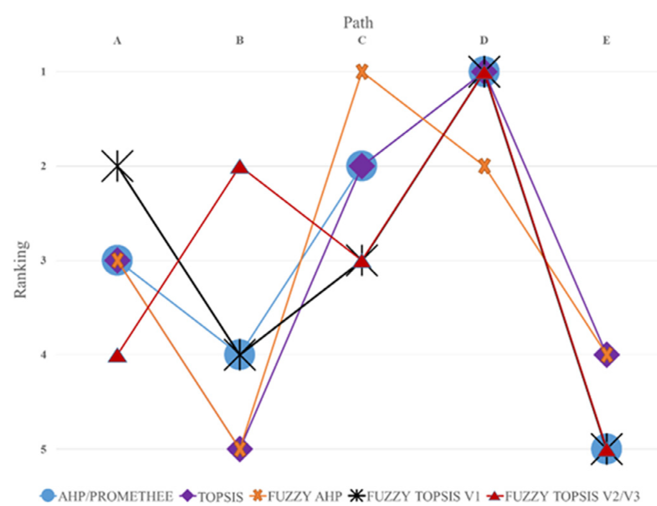

Fig. 33. Comparison of alternative path ranks assigned by different methods.

normalized again. In this matrix, higher numbers indicate more favorable results.

Table 13 presents the results of the decision analysis methods and the weights assigned to each alternative path.

Table 14 and Fig. 33 demonstrate the ranking results of the alternative paths using different multi-criteria decision analysis methods. The fastest path in the case study has a very high crash risk. Although the user indicated that the travel time is the most significant factor, almost all of the decision-making methods suggested path 4 (D) as the best compromise solution. Considering the high crash rate on the fastest path (B) and the longer travel time of the safest path (C), path 4 (D) is an appropriate choice.

The ranking results of AHP and PROMETHEE are identical. Similarly, the ranking results of fuzzy TOPSIS v1 and v2 that are calculated based on the fuzzification of the weights are identical. Excluding fuzzy AHP approach, all the seven methods selected 4 (alternative D) as the best compromised solution. For the evaluation, AHP is selected as the baseline due to being the most widely implemented method, the simplicity of calculation and the promising results. It is important to note that based on the weights of the alternatives assigned by AHP, the safest path (C) and path 4 (D) are almost equivalent. Consequently, fuzzy AHP which ranks the safest path as the first choice is acceptable and should not be considered as a huge negative point.

This case study and the user's judgement in this research are intentionally selected to demonstrate that despite the fact that the most important factor for the user is the travel time, the decision analysis methods should consider the extremely high probability of involving in an accident on the fastest path and ignore suggesting it. Nevertheless, the user has the option to select the fastest path manually or assign even a higher weight to the travel time to get a different result. Path 4 (D) is almost five times safer than the fastest path (B). TOPSIS v2 and v3 are the only methods that assigned a high ranking to the fastest path, which is inconsis-

Table 15
Sample average overlap calculation.

Depth (d)	S_d	T_d	Intersection	Overlap at depth	Average overlap
1	A	A	{A}	1/1 = 1	1/1 = 1
2	AB	AB	{AB}	2/2 = 1	(1+1)/2 = 1
3	ABC	ABC	{ABC}	3/3 = 1	(1+1+1)/3 = 1
4	ABCD	ABCE	{ABC}	3/4 = 0.75	(0.75+1+1+1)/4 = 0.9375
5	ABCDE	ABCED	{ABCDE}	5/5 = 1	(0.75+1+1+1+1)/5 = 0.95

Table 16
Average overlap results between AHP and the other methods.

Average overlap	Fuzzy AHP	TOPSIS	Fuzzy TOPSIS v1	Fuzzy TOPSIS v2	Fuzzy TOPSIS v3	PROMETHEE
AHP	0.75	0.95	0.90	0.83	0.83	1

tent with other decision-making methods. This can be the result of fuzzification of the crisp weights and incorporating these weights into the fuzzy TOPSIS algorithm. Fuzzy TOPSIS v1 which is based on the fuzzy weights obtained from fuzzy AHP, ranked the shortest path as the second-best compromise alternative path. Despite the fact that shortest path is considered as the safest path among the given alternatives, it should not be selected second due to the very long travel time.

Assuming that the Safe Route Planner only suggests one best compromise path, the results of all the decision-making methods are acceptable for the first case study. However, in order to suggest multiple paths and their rankings to the user, fuzzy TOPSIS v2 and v3 are undesirable. The ranking results of different methods must be compared against AHP to investigate the results of different decision-making methods. The focus is on the ranking results of alternatives instead of the actual weights of the alternatives due to the fact that two different decision making methods with different weights of alternatives might result in the same ranks or similar weights might result in different ranks.

The computed Spearman's rank correlation coefficient results between AHP and other methods are presented in Table 15. In comparison to AHP, both TOPSIS and fuzzy TOPSIS v1 switch one item pair but at different rank positions. TOPSIS switches path alternatives B and E at the bottom of the ranked list, while fuzzy TOPSIS switches path alternatives C and A in the middle of the list. Consequently, fuzzy TOPSIS v1 must be ranked lower than TOPSIS. The Spearsman's rank correlation coefficient fails and assigns similar weights to both TOPSIS and fuzzy TOPSIS v1.

Table 16 illustrates the average overlap results between AHP and the other multi-criteria decision analysis methods. PROMETHEE has an average overlap result of one meaning that its result is identical to the result of AHP. Obviously, the average overlap method fixed the problem with TOPSIS and Fuzzy TOPSIS v1 by assigning a higher rank to TOPSIS method. However, in this method, fuzzy AHP has the lowest score, since its first suggested path is different from other methods; therefore, it has the greatest impact. While fuzzy TOPSIS v2 and v3 scored higher than fuzzy AHP, their results are unacceptable since path B with a very high probability of accidents is ranked second. The result of Fuzzy AHP is more realistic than Fuzzy TOPSIS v2 and v3. In order to mitigate this issue, it is required to penalize a really bad path choice which appears at the top of the rank list.

Discounted Cumulative Gain (DCG) was applied as the last approach to overcome the aforementioned issues with Spearman's rank correlation coefficient and average overlap methods. The chosen relevance score for the Safe Route Planner with five unique alternative paths is {5, 4, 1, -4, -5}. The best alternative path will have a relevance score of 5, followed by the second-best alternative path with the relevance score of 4. Assigning a high rank by the decision-making analysis to a bad choice must be penalized,

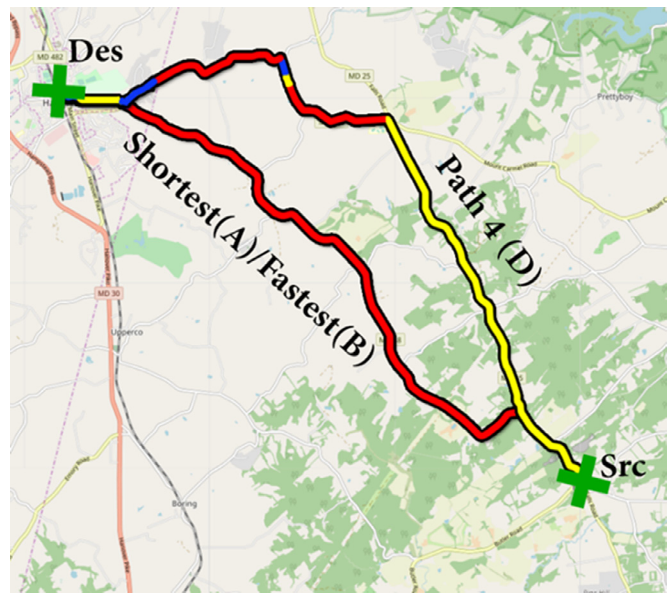

Fig. 34. Shortest, fastest, and Path 4 from Butler Road (Src) to Shiloh Avenue (Des).

for this reason, negative scores -4 and -5 are selected for the worst alternative paths. Since the AHP ranking is considered as the baseline or ideal solution, the relevance scores of the alternatives are assigned as *{D = 5, C = 4, A = 1, B = -4, E = -5}*. In addition, the base of the logarithm is selected to be three. The ranked list must be turned into gained value list (*G*) by replacing the alternative paths by their relevance scores, for instance the ranked list obtained from fuzzy AHP = {*C, D, A, E, B*} is turned into G = {4, 5, 1, -5, -4}.

Table 17 presents the discounted cumulative gain and the normalized DCG for all the decision-making methods used in the first case study. DCG was successful in evaluating the decision-making methods used for the Safe Route Planner. It assigned a lower rank to fuzzy TOPSIS v2 and v3 than to fuzzy AHP. In addition, DCG clearly considered the top of the ranked list more important than the bottom of the list and penalized fuzzy TOPSIS v1. Consequently, it assigned a higher score to TOPSIS and a lower score to fuzzy TOPSIS v1. Finally, DCG assigned a higher score to fuzzy AHP than to fuzzy TOPSIS v1. By assigning a higher relevance score to the first and second best paths, it is possible to punish fuzzy TOPSIS v1 further for choosing the shortest path with a long travel time as the second best alternative.

The second evaluation is based on a use case where the user intends to travel from the intersection of Falls Road and Butler Road, Butler in Maryland to the intersection of Shiloh Avenue and Main Street, Hampstead in Maryland as shown in Figs. 34 and 35. The

Table 17
Discounted cumulative gain and normalized DCG results between AHP and the other methods.

	AHP	Fuzzy AHP	TOPSIS	Fuzzy TOPSIS v1	Fuzzy TOPSIS v2	Fuzzy TOPSIS v3	PROMETHEE
DCG	6.921	6.267	6.851	6.299	4.711	4.711	6.921
nDCG	1	0.906	0.990	0.910	0.681	0.681	1

Table 18
Comparison of alternative path weights assigned by different methods.

Path	AHP	Fuzzy AHP	TOPSIS	Fuzzy TOPSIS V1	Fuzzy TOPSIS V2	Fuzzy TOPSIS V3	PROMETHEE
Shortest (A)	0.2048	0.1970	0.2159	**0.2389**	**0.2555**	**0.2412**	0.2012
Fastest (B)	0.2048	0.1970	0.2159	**0.2389**	**0.2555**	**0.2412**	0.2012
Safest (C)	**0.2071**	**0.2287**	0.1772	0.1424	0.0759	0.1141	**0.2018**
Path 4 (D)	0.2062	0.2094	**0.2518**	0.2327	0.2253	0.2303	0.2016
Path 5 (E)	0.1770	0.1679	0.1394	0.1471	0.1878	0.1733	0.1943

Table 19
Comparison of alternative path ranks assigned by different methods.

Path	AHP	Fuzzy AHP	TOPSIS	Fuzzy TOPSIS V1	Fuzzy TOPSIS V2	Fuzzy TOPSIS V3	PROMETHEE
Shortest/Fastest (A/B)	3	3	2	1	1	1	3
Safest (C)	1	1	3	4	4	4	1
Path 4 (D)	2	2	1	2	2	2	2
Path 5 (E)	4	4	4	3	3	3	4

Table 20
Discounted cumulative gain and normalized DCG results between AHP and the other methods.

	AHP	Fuzzy AHP	TOPSIS	Fuzzy TOPSIS v1	Fuzzy TOPSIS v2	Fuzzy TOPSIS v3	PROMETHEE
DCG	5.34	5.34	2.89	−2.89	−2.89	−2.89	5.34
nDCG	1	1	0.54	−0.54	−0.54	−0.54	1

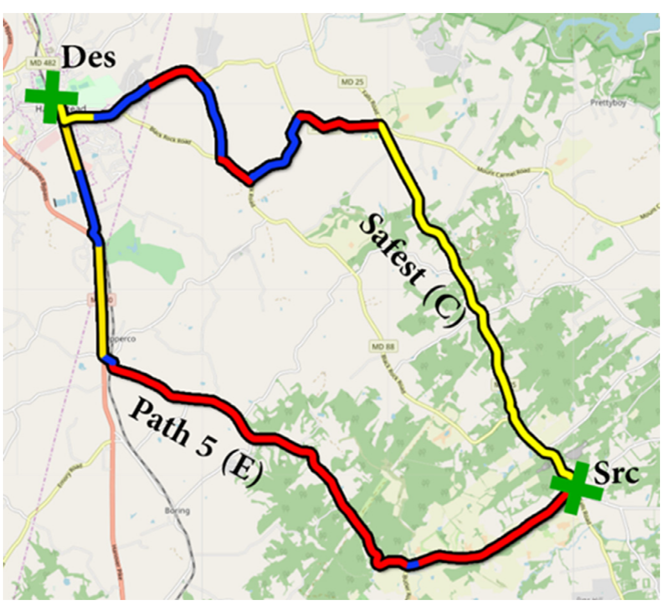

Fig. 35. Safest and Path 5 from Butler Road (Src) to Shiloh Avenue (Des).

$$\begin{array}{rc} & \begin{array}{ccc}\text{Safety} & \text{Distance} & \text{Time}\end{array} \\ \begin{array}{r}\text{Shortest }(A)\\ \text{Fastest }(B)\\ \text{Safest }(C)\\ \text{Path 4 }(D)\\ \text{Path 5 }(E)\end{array} & \begin{bmatrix} 181.7542 & 14.8009 & 715 \\ 181.7542 & 14.8009 & 715 \\ 90.0118 & 18.1377 & 1009 \\ 133.2662 & 16.5245 & 790 \\ 236.8785 & 17.4077 & 797 \end{bmatrix}\end{array}$$

Fig. 36. Five alternative paths and their properties.

matrix shown in Fig. 36 presents the alternative paths, the safety level, the distance and the travel time for each alternative path. In this illustration, the fastest and shortest paths are the same. Except the safest path, all other paths are very risky due to a very high crash rate. Path 5 (E) has the worst safety level, and its travel time is almost similar to the shortest and fastest path; therefore, it must be considered as the last suggested path.

Table 18 presents the results of the decision analysis methods and the weights assigned to each alternative path.

Table 19 demonstrates the ranking results of the alternative paths in the second case. The ranking results of AHP, fuzzy AHP, and PROMETHEE are identical. Similarly, the ranking results of fuzzy TOPSIS v1, v2 and v3 are identical. There is a little difference between the safest path (C) and path 4 (D); therefore, both of them can be accepted to be the first suggested path. Fuzzy TOPSIS (v1, v2, and v3) suggested the shortest path and the fastest path as the best paths; however, due to the high crash rate, they should not be considered as the best option. Path 5 (E) has the worst safety level, and its travel time and distance are longer than alternatives A, B, and D. In addition, although path 5 (alternative E) is 26% faster than the safest path (alternative C), the chances of involving in an accident is 2.6 times higher on this path than the safest path. As a result, path 5 (alternative E) must be considered as the last suggested path. All the decision-making methods except Fuzzy TOPSIS (v1, v2, and v3) considered path 5 (alternative E) as the last suggested path.

Table 20 presents the discounted cumulative gain and the normalized DCG for all the decision-making methods used in the second case study. For this evaluation, the duplicates in the results must be removed; therefore, alternative *B* were removed and only four different alternative paths were presented. The chosen relevance score for the Safe Route Planner with four unique alternative paths is {5, 4, -4, -5}.

14. Conclusion

In this paper, the authors presented the addition of a decision-making system, AHP, to the Safe Route Planner. Using Analytical Hierarchy Process, the user can now specify his or her preferences in regards to safety level, travel time and distance. The navigation system first finds multiple paths, analyzes them and suggests the best possible option for each user. The addition of this MCDM method led to a comparative study of decision-making methods.

There are a few comparative studies to compare the results of different MCDA methods, especially in the transportation field. Another novelty of this paper is the unique selection and comparison of five different decision-making methods in the transportation field using two real world case studies. Based on the results, the PROMETHEE ranking fits well with the AHP ranking, and they both produce the best results. Although fuzzy AHP is the only decision-making method that suggested a different best compromise solution in the first case study, its result is considered acceptable. TOPSIS and Fuzzy TOPSIS produce poor ranking results; the initial investigation showed that this can be attributed to their normalization methods. However, further investigation is needed to validate this finding.

Among the five MCDM methods studied in this research, we suggest using AHP as a simple and robust method in the transportation field. Although PROMETHEE produces identical results to AHP, it requires the weights of the relative importance of the criteria to be known; therefore, it depends on a method such as AHP to derive the criteria weights. Furthermore, the final weights of the alternatives (net flow) in PROMETHEE ranges between minus one and one which is confusing for the decision makers to interpret; therefore, the range should be changed to [0, 1] using the method presented in the PROMETHEE section.

Another contribution of this study is exploring three ways to derive criteria weights and feeding them into fuzzy TOPSIS method. The findings suggest that using fuzzy AHP to derive triangular fuzzy weights of criteria and directly using these criteria weights to construct the weighted normalized fuzzy decision matrix in the fuzzy TOPSIS method yields better results. In contrast, fuzzification of the crisp weights using linguistic terms and feeding them into the fuzzy TOPSIS method produce inconsistent results.

Furthermore, comparing the rank list results of Safe Route Planner is unique in the sense that the similarity measurement algorithms need to consider that the top of the list is more important than the bottom. In addition, it is required to penalize a really bad path choice that appears at the top of the rank list. We applied different rank evaluation methods and found that Discounted Cumulated Gain which is used in information retrieval evaluation is a suitable evaluation method.

The limitations of the current study are as follows:

- In the road safety calculation, the time of accidents should be considered to have different weights. For instance, an accident that occurred 5 years ago must have less impact on the safety level than an accident that occurred a month ago.
- It is necessary to add a new user interface to allow a group of experts to use their judgments and assign weights to different types of accidents, namely, fatal, injury, property damage, and real-time accidents. This will allow a better calculation of the weighted crash rate.
- The relevance scores for the evaluation performed by Discounted Cumulated Gain were selected manually. In the case of a tie in the ranked lists, it is required to revise the relevance scores. This resulted in two different relevance scores for the two different case studies. Hence, a more robust and automated approach is needed to find and assign the relevance scores.
- The fuzzy TOPSIS method presented in this research is based on Chen's (2000) fuzzy TOPSIS method which is widely used. There are different methods of fuzzy TOPSIS available in the literature, it is important to evaluate other methods of fuzzy TOPSIS to investigate the weaknesses of and strength of Chen's fuzzy TOPSIS.

For future work, we aim to address the followings:

- A possible area of work is the addition of backward chaining expert system to the Safe Route Planner to find out why drivers have a specific home-to-work path. First, a user study needs to be performed to identify the daily path users take from home to work. Next, users' preferences based on safety, distance and time criteria must be calculated. Finally, backward chaining expert system will be used to work backward through inference rules and find the facts that resulted in the path selection for individuals and compare their daily paths against the suggested paths by the Safe Route Planner.
- Another future work will be the addition of an interface for the field experts to evaluate and rank road segments based on their knowledge, and an inference module to combine the ranking of experts and the factual knowledge to derive the best answer.
- The system should be able to aggregate users' preference inputs through the user interface, incorporate them to the knowledge base, and train itself to better predict users' behavior and what criteria they are more interested in.
- Another future work will be to conduct a user study that consists of three phases. First, we will generate a questionnaire to evaluate the perception of the users on the risk of driving on different roads and if they are willing to take a safer path over a faster path. This phase should be conducted for different age groups, novice drivers and experienced drivers. Phase two of the user study will be to demonstrate the risk level of some sample road networks, educating the users on how to use the Safe Route Planner. In phase three, we will get users' feedback using a similar questioner generated on phase one and compare the results.

Declaration of Competing Interest

We confirm that this work is original and has not been published elsewhere, nor is it currently under consideration for publication at another journal. We have no conflicts of interest to disclose.

Credit authorship contribution statement

Reza Sarraf: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing - original draft. **Michael P. McGuire:** Writing - review & editing, Supervision.

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